L02 — study notes

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L02 — Topic 2A · Valuation, the NPV rule and the frictionless benchmark · Map

PDF. L02-notes.pdf — this whole unit in one printable file (rendered from these notes; rebuild with `python -I ~/.claude/commands/templates/notes_pdf.py Notas/L02` after any edit).

One claim. Value is a present value at a risk-adjusted rate: discount the unlevered free cash flows at the return that the assets' systematic risk commands. That rate comes from comparables only after **unlevering**, because $A = D + E$ makes asset returns and betas value-weighted averages of debt's and equity's. **The NPV rule is the shareholders' rule** when markets are frictionless. **In that benchmark financing is irrelevant:** with complete markets every claim is priced state by state, so slicing the same payoff into debt and equity cannot change its total value (MM I). The cost of equity rises exactly enough to keep the WACC fixed (MM II).

Layer	What it means for this unit
U — undergraduate	Ross, Higgins, Berk: the SML, FCF rules, NPV vs. IRR, Polaris' unlevering, Transmadruga and the coffee shop, Ricardo S/A
C — course (examinable)	the slides: (1.1)–(1.12), the comparables computation, FCF (2.1), NPV = shareholders' gain, IRR pitfalls, payoffs as options, state prices by replication, the entrepreneur tables, both MM I proofs, MM II, $y > r_D$
G — graduate	Amaro de Matos' Fisher separation (Prop. 1.1.1), no-arbitrage ⇔ positive state prices (Prop. 1.2.1), MM1/MM2 with state prices; Tirole's risk-neutral proof. **Named, not examinable**

Reading order

Part I — valuing an asset (2A.1–2A.4)

1. L02-01-present-value-capm-betas — slides 2–23: P_A as an expected PV, the CAPM rate, **why a levered comparable's equity beta overstates the assets'** (the house: β_E 1.25 and 2), **(1.10)–(1.11) derived**, and the **three comparables → 10.56%**.
2. L02-02-cash-flows-npv-irr — slides 24–38: unlevered FCF and why the tax shield stays out, incremental/sunk/opportunity, terminal value derived; **NPV = shareholders' gain = 5 under three financings**; the IRR's four failures (two roots, none, lending vs. borrowing, scale).

Part II — the claims and the benchmark (2A.5–2A.6)

3. L02-03-debt-vs-equity-claims — slides 39–45: four differences, covenants, limited liability; $\min(\tilde{A}, F) + \max(\tilde{A} - F, 0) = \tilde{A}$; **equity is a call**, debt a bond minus a put.
4. L02-04-complete-markets-state-prices — slides 46–54: the assumptions and **what each buys**; contingent claims, completeness, the two arbitrages; **state prices 0.381 / 0.571 by replication**; why they are not probabilities.

Part III — irrelevance (2A.7–2A.10)

5. L02-05-entrepreneur-example — slides 55–81: every table. $p_1 = 0.357$, $p_2 = 0.595$; $A = 1.75$ **and NPV 0.75 under debt, equity ($\alpha = 4/7$) and any mix**; $r_A = 20\%$ fixed, r_E 24.4% → 26% → 40%; **the 1.77 fallacy**; slide 81's exercise solved.
6. L02-06-mm1-arbitrage-put-call — slides 82–96: **MM I by arbitrage in both directions** (undoing leverage, homemade leverage), risky debt as bond − put, **put–call parity ⇒ $V(K)$ constant**, checked at $K = 1.05$ and 2.1.
7. L02-07-mm2-wacc-cost-of-debt — slides 97–105: **MM II derived**, the exact derivative and **the flat 40% region** once debt is risky, $r_{WACC} = r_A = r_U$, conservation of value, and $y = 52.73\% > r_D = 14.55\%$.

Support

8. L02-08-formulas — the block list as exam specification; every equation; every number.
9. L02-09-pitfalls — ten confusions (two slide displays among them) and 25 True/False items.
10. L02-10-python — reference implementation; every slide number as an `assert`.

The one result to hold

$$A = \sum_s p_s X_s = \underbrace{\sum_s p_s \min(X_s, F)}_{D} + \underbrace{\sum_s p_s \max(X_s - F, 0)}_{E} \quad \text{for every } F \implies r_E = r_A + \frac{D}{E}(r_A - r_D)$$

Pricing is linear in payoffs and the two claims add up to the asset: that is MM I. MM II is the same statement in returns.

Numbers to have memorized

Quantity	Value	Where
Project cost of capital from comparables	$\bar{\beta}_A = 0.78$, $r = 10.56\%$	slides 22–23
Entrepreneur state prices	$p_1 = 1/2.8 = 0.357$ (boom), $p_2 = 0.595$ (slump)	slides 65–66
Entrepreneur firm	$A = 1.75$, NPV 0.75, $\alpha = 4/7$, $r_A = 20\%$	slides 68–73
r_E at $D = 0,\ 0.4,\ 0.5,\ 1$	20%, 24.4%, 26%, 40%	slides 71–80
NPV = shareholders' gain	5 (equity 50 → 70 less 15; $\alpha = 15/70$; $F = 16.5$)	slides 32–36

Figures

Built by `figures/make_figures.py` (run: `python -I Notas/L02/figures/make_figures.py`; it ends with asserts on the plotted numbers).

Figure	Shows	Lives in
L02-sml-comparables	the SML; three comparables moved from β_E to β_A; the project at 0.78 / 10.56%	L02-01-present-value-capm-betas §6
L02-npv-profile	NPV(r) for Ross' mine (IRRs 25%, 33⅓%) and slide 38's lending vs. borrowing	L02-02-cash-flows-npv-irr §5
L02-payoffs	$\min(\tilde{A}, F)$, $\max(\tilde{A} - F, 0)$ vs. $\tilde{A}$, $F = 1.05$, the project's two outcomes	L02-03-debt-vs-equity-claims §2
L02-state-prices	the state-price plane: stock and bond on the price-1 line, project on 1.75	L02-05-entrepreneur-example §7
L02-mm2	r_E, r_D, r_A vs. D/E: linear while safe, flat at 40% when risky	L02-05-entrepreneur-example §7, L02-07-mm2-wacc-cost-of-debt §2

Interactive companions

- The State-Price Balance Sheet (`companions/companion-mm-state-prices.html`) — controls: boom and slump payoffs, the two state prices, the face value of debt F; read-outs A (the answer), D, E, NPV, r_D (with the yield), r_E (with D/E), r_A, r_f, and put–call parity live; presets for slides 68–80 and a risky $F = 2.1$. Linked from notes 03–07. Checked by `companions/check_companions.py` (node runs the page's own model at six control vectors).

Scope and ceiling

- **Examinable:** the slides' content (C layer) above, at the slides' level: one-period, finitely many states, CAPM rates, no taxes.

- **Named only (G):** Fisher separation, Farkas/positive state prices, incomplete markets' non-unique prices, MM with state prices (Amaro de Matos); Tirole's risk-neutral proof and payout irrelevance.
- **Out of this unit:** taxes, distress costs and the trade-off (Topic 2B, L03); debt overhang, risk-shifting and asymmetric information are flagged where the slides flag them and left to the agency and information topics of the syllabus.

Course-wide position

Unit	Relation to L02
L01 — Topic 1, accounting review	supplies $A = D + E$ in market value and the FCF formula; ROE and P/E rising with leverage meet MM II here
L02 — Topic 2A	**present value, CAPM rates, NPV, state prices, MM I and II without frictions**
L03 — Topic 2B, taxes and trade-off	relaxes assumption 3: $V_L = V_U + \tau D$, after-tax WACC, APV, distress costs, the static trade-off

Rules packs: rules/02 — valuation · rules/03 — MM benchmark. **Block map:** Map/lectures-index (Topic 2A) · Map/study-guide.

Bibliography actually used

Book	Where
Ricca, Topic 2A slides — the spine	every note
Ross, *Fundamentos* 9e, Cap. 9 (pp. 282, 297–298), Cap. 10 §10.2 (p. 320), Cap. 13 §13.7 (pp. 442, 447), Cap. 14 §14.5 (pp. 476–477), Cap. 16 §16.2–16.3 (pp. 543, 545–549)	U layer of notes 01, 02, 06, 07
Higgins (Koski–Higgins–Mitton) 13e, Ch. 7 (pp. 246, 248, 258, 263), Ch. 8 (pp. 302–303) and App. (pp. 318–319)	notes 01, 02
Berk–DeMarzo–Harford, *Fundamentals* 2e, Ch. 8 (pp. 213, 221–223), Ch. 9 (pp. 258–259), Ch. 12 (p. 368), Ch. 13 (pp. 384, 396–398), Ch. 16 §16.2 (pp. 464–471)	notes 01, 02, 05, 06, 07
Amaro de Matos, Ch. 1 §1.1.5–1.1.6 (pp. 11–14), §1.2.1–1.2.4 (pp. 18, 21–25); Ch. 2 §2.1.1–2.1.2 (pp. 40–43)	G layer of notes 02, 03, 04, 06, 07
Tirole, Ch. 2 §2.2 (pp. 77–78)	notes 03, 06

Related project material

- Lists: lista-r2 — valuation (solutions PDF, checks `Listas/lista-r2-valuation-solutions_check.py`, roadmap roteiro-r2, pack study-pack-r2); lista-r3 — MM, taxes, trade-off (solutions PDF, checks `Listas/lista-r3-mm-taxes-tradeoff-solutions_check.py`, roadmap roteiro-r3, pack study-pack-r3).

- Problem sets: PS1 Ex. 2 (p. 1), Ex. 3 (pp. 1–2); PS2 Ex. 1(a–e) (p. 1), Ex. 3(a–b) (p. 2), Ex. 4 (pp. 2–3). PS2 Ex. 1(f) and Ex. 3(c) belong to Topic 2B.
- Quiz: quiz-topic2a-valuation-mm.
- Audio prompt: topic2a-audio.
- Older notes on Tirole Ch. 1–2 from the syllabus phase: aula-01-notes (related, not this lecture).

2A.1–2A.2 · Present value, the CAPM discount rate, and unlevering a comparable's beta

> **Claim.** An asset is worth its expected cash flows discounted at the return investors require for bearing its **systematic** risk. For a project there is no price history, so its beta is borrowed from a traded firm with similar assets. That firm's observable **equity** beta mixes business risk with leverage, so it is first **unlevered**: because $A = D + E$ in market value, returns and betas of the assets are value-weighted averages of those of debt and equity.

Prev: — · Next: L02-02-cash-flows-npv-irr · Map: L02-00-map

Notation.

Symbol	Meaning	Range
P_A	price (value) at $t = 0$ of asset/project A	$\mathbb{R}$
$\tilde{d}_t$	random cash flow at date $t = 1, \dots, T$ ($T \to \infty$ allowed)	$\mathbb{R}$
r_t	expected return (discount rate) for a date-t cash flow; r if constant	> -1
$\tilde{r}$, $r = E[\tilde{r}]$	realized and expected return	—
r_f	risk-free rate	> -1
$\tilde{r}_m$, $E[\tilde{r}_m]$	market return and its mean	—
$\beta = \mathrm{Cov}(\tilde{r}, \tilde{r}_m)/\mathrm{Var}(\tilde{r}_m)$	systematic risk	$\mathbb{R}$
A, D, E	**market** values of assets, debt, equity of a firm, $A = D + E$	≥ 0
$\tilde{C}_A, \tilde{C}_D, \tilde{C}_E$	cash flows at $t = 1$ to assets, debt, equity, $\tilde{C}_A = \tilde{C}_D + \tilde{C}_E$	—
r_A, r_D, r_E; $\beta_A, \beta_D, \beta_E$	expected returns and betas of assets, debt, equity	—
C, F, N	bond coupon, face value, maturity	≥ 0

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1 · C — The price of an asset is a present value

The asset generates a random stream $\tilde{d}_1, \dots, \tilde{d}_T$ (slide 2). Its price is the expected present value (slide 4):

$$P_A = \sum_{t=1}^{T} \frac{E[\tilde{d}_t]}{(1 + r_t)^t} \tag{1.1}$$

- $r_t = E[\tilde{r}_t]$ is an **expected** return. The **realized** return $\tilde{r}_t$ is random because the $\tilde{d}_t$ are, so ex post it can land above or below r_t (slide 4).
- The value is computed **regardless of how the asset is financed** (slide 3): (1.1) prices the asset, not any claim on it. Why that is legitimate is exactly MM I (L02-06-mm1-arbitrage-put-call).

Riskless case (slide 5). With r_t the t-year spot rate, a coupon bond paying C at $t = 1, \dots, N$ and F at N is (1.1) with $E[\tilde{d}_t] = d_t$ known:

$$P = \frac{C}{1+r_1} + \frac{C}{(1+r_2)^2} + \cdots + \frac{C}{(1+r_{N-1})^{N-1}} + \frac{C+F}{(1+r_N)^N} \tag{1.2}$$

Illustration (numbers chosen here, not on the slides): $N = 2$, $C = 5$, $F = 100$, $r_1 = 4\%$, $r_2 = 5\%$. Substituting into (1.2):

$$P = \frac{5}{1.04} + \frac{5+100}{(1.05)^2}$$

$$P = 4.8077 + \frac{105}{1.1025} = 4.8077 + 95.2381 = 100.0458$$

With a **time-invariant** rate $r_t = r$ (slide 6), (1.1) becomes

$$P_A = \sum_{t=1}^{T} \frac{E[\tilde{d}_t]}{(1+r)^t}. \tag{1.3}$$

The two hard inputs are the forecasts $E[\tilde{d}_t]$ (L02-02-cash-flows-npv-irr) and the rate r (this note). Slide 7:

$$\underbrace{r}_{\text{required expected return}} = \underbrace{r_f}_{\text{compensation for delaying consumption}} + \underbrace{\text{risk premium}}_{\text{compensation for bearing risk}}$$

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2 · C — The CAPM rate

The risk premium comes from an asset-pricing model; the course uses the CAPM (slide 8):

$$r = E[\tilde{r}] = r_f + \beta\,(E[\tilde{r}_m] - r_f) \tag{1.4}$$

$$\beta = \frac{\mathrm{Cov}(\tilde{r}, \tilde{r}_m)}{\mathrm{Var}(\tilde{r}_m)}, \qquad \mathrm{Cov}(X,Y) = E\big[(X - E[X])(Y - E[Y])\big] = E[XY] - E[X]E[Y] \tag{1.5}$$

> **Only the market-covariance part of risk is priced.** β is the quantity of systematic risk; $E[\tilde{r}_m] - r_f$ is its price, common to every asset. (Ross §13.7, p. 442 builds the same line from portfolios of a risky asset and the risk-free asset; Berk p. 368 quotes Sharpe: *"expected return is proportionate to beta relative to the market portfolio."*)

Estimation (slide 9). r_f: government bond yields at a maturity representative of the project's duration. $E[\tilde{r}_m] - r_f$: slow-moving consensus values. β: from past returns, for traded securities. **A project has no past returns** — hence comparables.

U-layer example (Higgins pp. 302–303, his Eq. 8.2): Polaris' equity beta 2.05, $r_f = 1.7\%$, market premium 6.4%:

$$K_E = 1.7\% + 2.05 \times 6.4\%$$

$$K_E = 1.7\% + 13.12\% = 14.82\% \approx 14.8\%$$

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3 · C — Comparables: borrow the beta of similar assets

If a traded firm has assets with systematic risk β_A similar to the project's, then the project's beta is $\beta = \beta_A$ and, substituting into (1.4) (slide 10):

$$r = r_f + \beta_A(E[\tilde{r}_m] - r_f). \tag{1.6}$$

The problem (slide 11): only the comparable's **equity** beta β_E is estimable from stock prices. β_D is estimable only if its debt trades — usually it does not (bank debt, trade credit, illiquid bonds); fall-backs are bond indices of the same rating, or $\beta_D = 0$ when leverage is low and default very unlikely.

> **Ross §14.5 (pp. 476–477) shows why one rate for everything fails:** with the firm's WACC of 15% as a single hurdle, a project of beta 0.60 expected to earn 14% is rejected, although its required return is only $7\% + 0.60 \times 8\% = 11.8\%$. One hurdle rate accepts too many risky projects and rejects too many safe ones; a **project-specific** beta fixes it (Berk §13.5, pp. 396–398, does the same with a pure-play comparable).

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4 · C — Leverage raises equity risk: the house (slides 12–15)

A house worth 100 moves one-for-one with the market. The market return is +10% with probability 0.8 and −5% with probability 0.2; $r_f = 2\%$.

$$E[\tilde{r}_m] = 0.8(0.10) + 0.2(-0.05) = 0.08 - 0.01 = 0.07 \tag{1.7}$$

$$\mathrm{Var}(\tilde{r}_m) = 0.8(0.10 - 0.07)^2 + 0.2(-0.05 - 0.07)^2 = 0.8(0.0009) + 0.2(0.0144) = 0.0036 \tag{1.8}$$

So $r_A = r_m = 7\%$ and $\beta_A = 1$; tomorrow the house is worth $100(1.10) = 110$ or $100(0.95) = 95$.

Case 1 — 20% debt. Debt $D = 20$ is riskless: it pays $20(1.02) = 20.4$ in both states. Equity $E = 80$ gets the residual: $110 - 20.4 = 89.6$ (up) and $95 - 20.4 = 74.6$ (down).

	Today	Up	Down	Return up	Return down	Mean	Beta
A	100	110	95	10%	−5%	7%	1
D	20	20.4	20.4	2%	2%	2%	0
E	80	89.6	74.6	12%	−6.75%	**8.25%**	**1.25**

Equity returns: $89.6/80 - 1 = 0.12$ and $74.6/80 - 1 = -0.0675$. Then

$$E[\tilde{r}_E] = 0.8(0.12) + 0.2(-0.0675) = 0.096 - 0.0135 = 0.0825$$

$$\mathrm{Cov}(\tilde{r}_E, \tilde{r}_m) = 0.8(0.12 - 0.0825)(0.03) + 0.2(-0.0675 - 0.0825)(-0.12) = 0.0009 + 0.0036 = 0.0045$$

$$\beta_E = \frac{0.0045}{0.0036} = 1.25$$

> **The slide prints the down return as −7% and the mean as 8%** (slide 14): both are rounded from −6.75% and 8.25%. The beta, 1.25, is exact. Check with (1.12) below: $r_E = 7\% + \frac{20}{80}(7\% - 2\%) = 8.25\%$.

Case 2 — 50% debt. $D = 50$ pays 51; equity 50 gets 59 or 44, returns $+18\%$ and -12%:

$$E[\tilde{r}_E] = 0.8(0.18) + 0.2(-0.12) = 0.144 - 0.024 = 0.12$$
$$\mathrm{Cov}(\tilde{r}_E, \tilde{r}_m) = 0.8(0.06)(0.03) + 0.2(-0.24)(-0.12) = 0.00144 + 0.00576 = 0.0072$$
$$\beta_E = \frac{0.0072}{0.0036} = 2$$

In both cases $r_E > r_A$ and $\beta_E > \beta_A$ (slide 15). The equity of a highly levered firm **overstates** the risk of its assets (slide 12). The house did not change; only the claim on it did.

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5 · C — Unlevering: asset returns and betas are value-weighted averages

First derivation (slide 17). All cash generated by the assets goes to debt- or equityholders, so investing 1 in the assets is investing $D/(D+E)$ in the debt and $E/(D+E)$ in the equity.

Second derivation (slides 18–19). One-period firm. Each value is its expected payoff discounted at its own expected return:

$$D = \frac{E[\tilde{C}_D]}{1+r_D}, \qquad E = \frac{E[\tilde{C}_E]}{1+r_E}, \qquad A = \frac{E[\tilde{C}_A]}{1+r_A} \tag{1.9}$$

Take A from (1.9) and use $\tilde{C}_A = \tilde{C}_E + \tilde{C}_D$ (linearity of $E[\cdot]$):

$$A = \frac{E[\tilde{C}_E] + E[\tilde{C}_D]}{1+r_A}$$

Substitute $E[\tilde{C}_E] = E(1+r_E)$ and $E[\tilde{C}_D] = D(1+r_D)$, both from (1.9):

$$A = \frac{(E(1+r_E)) + (D(1+r_D))}{1+r_A}$$

Multiply by $(1+r_A)/A$:

$$1 + r_A = \frac{E}{A}(1+r_E) + \frac{D}{A}(1+r_D)$$

Use $A = D + E$, so $E/A + D/A = 1$, and subtract 1 from both sides:

$$\boxed{r_A = \frac{E}{D+E}\, r_E + \frac{D}{D+E}\, r_D} \tag{1.10}$$

Realized returns (slide 20). The same steps with $D = \tilde{C}_D/(1+\tilde{r}_D)$, etc., give $\tilde{r}_A = \frac{E}{D+E}\tilde{r}_E + \frac{D}{D+E}\tilde{r}_D$ state by state. The weights $D/(D+E)$ and $E/(D+E)$ are **today's values — constants**.

Betas (slide 21). Covariance is linear and the weights are constants:

$$\mathrm{Cov}(\tilde{r}_A, \tilde{r}_m) = \mathrm{Cov}\Big(\tfrac{E}{D+E}\tilde{r}_E + \tfrac{D}{D+E}\tilde{r}_D,\ \tilde{r}_m\Big) = \tfrac{E}{D+E}\,\mathrm{Cov}(\tilde{r}_E, \tilde{r}_m) + \tfrac{D}{D+E}\,\mathrm{Cov}(\tilde{r}_D, \tilde{r}_m)$$

Divide by $\mathrm{Var}(\tilde{r}_m)$ and use (1.5) term by term:

$$\boxed{\beta_A = \frac{E}{D+E}\,\beta_E + \frac{D}{D+E}\,\beta_D} \tag{1.11}$$

Solving (1.10) and (1.11) for the equity side gives the **relevering** forms used later:

$$r_E = r_A + \frac{D}{E}(r_A - r_D), \qquad \beta_E = \beta_A + \frac{D}{E}(\beta_A - \beta_D) \tag{1.12}$$

(multiply (1.10) by $(D+E)/E$: $\frac{D+E}{E}r_A = r_E + \frac{D}{E}r_D$, then write $\frac{D+E}{E}r_A = r_A + \frac{D}{E}r_A$). **With** $\beta_D = 0$, (1.11) is $\beta_A = \frac{E}{D+E}\beta_E$ (slide 23; Higgins p. 319, who adds that $\beta_A = \beta_E$ for an all-equity firm and that β_E rises above β_A as leverage grows).

> **Which assumption drives (1.10)–(1.11)?** Only $A = D + E$ in **market value** and $\tilde{C}_A = \tilde{C}_D + \tilde{C}_E$ — no CAPM, no MM. They are accounting identities for market values. MM I adds that A (hence r_A) does not move when D/E moves (L02-07-mm2-wacc-cost-of-debt).

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6 · C — Worked example: three comparables (slides 22–23)

$r_f = 4\%$, $E[\tilde{r}_m] = 12.4\%$; the comparables' debt is risk-free ($\beta_D = 0$); book debt proxies market debt; all three are equally good comparables.

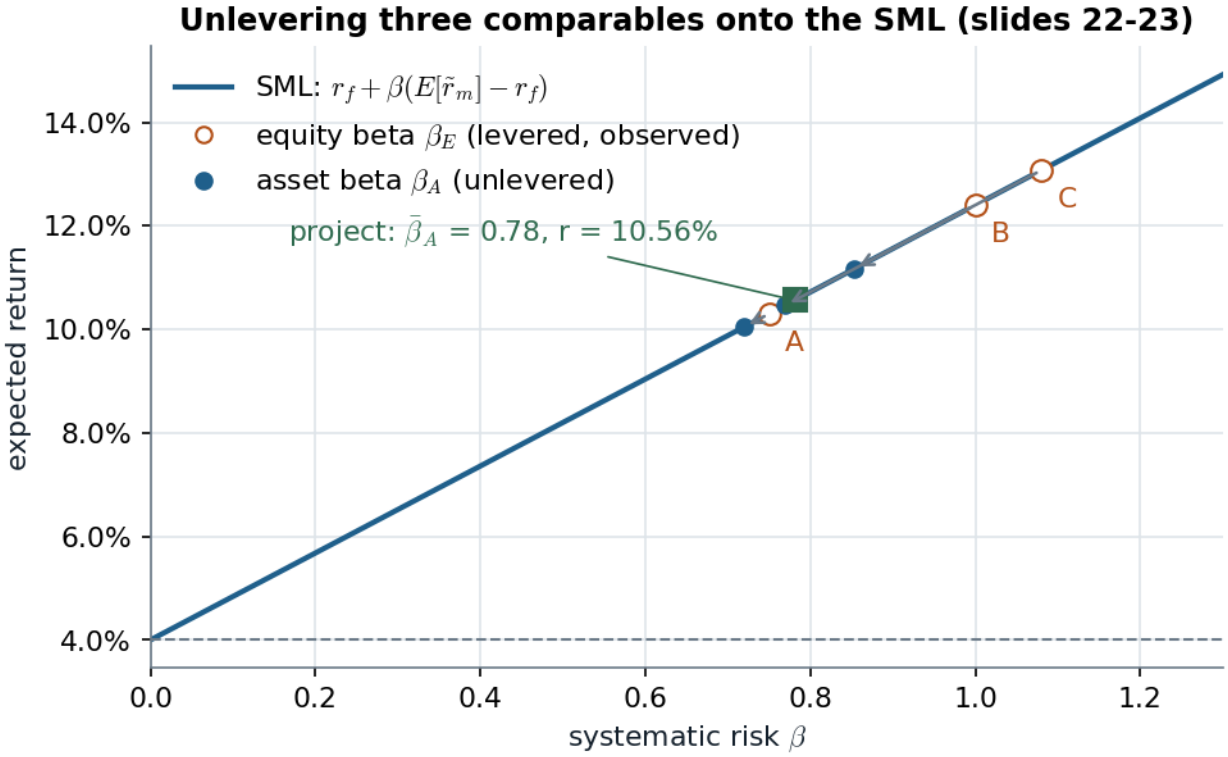
Firm	β_E	D	E	$E/(D+E)$	$\beta_A = \frac{E}{D+E}\beta_E$
A	0.75	0.004	0.096	$0.096/0.100 = 0.96$	$0.96 \times 0.75 = 0.72$
B	1.00	2.3	7.7	$7.7/10.0 = 0.77$	$0.77 \times 1.00 = 0.77$
C	1.08	0.21	0.79	$0.79/1.00 = 0.79$	$0.79 \times 1.08 = 0.8532$
Average					$(0.72 + 0.77 + 0.8532)/3 = 0.7811$

Substituting $\bar{\beta}_A = 0.7811$ into (1.6):

$$r = 4\% + 0.7811\,(12.4\% - 4\%)$$

$$r = 4\% + 0.7811 \times 8.4\% = 4\% + 6.561\% = 10.56\%$$

> **The slide shows 0.78 but uses the unrounded 0.7811:** $0.78 \times 8.4\% = 6.552\%$ would give 10.55%. Average the **asset** betas, never the equity betas: the equity-beta average, $(0.75 + 1 + 1.08)/3 = 0.943$, would charge $4\% + 0.943 \times 8.4\% = 11.92\%$.



Unlevering moves each comparable down the SML; the project sits at the average asset beta (eq. 1.6, 1.11; slides 22–23).

Each arrow is the leverage premium the equity beta carried. Firm A is almost all equity, so it barely moves; B and C move most because their debt is 21–23% of value. The project's rate is read at the green square, below B's and C's equity points.

U-layer check (Higgins p. 319). Polaris: $E = 5{,}898$, $V = D + E = 7{,}334$ (millions), $\beta_E = 2.05$:

$$\beta_A = \frac{5{,}898}{7{,}334} \times 2.05 = 0.80420 \times 2.05 = 1.65$$

---

7 · G — Risky debt in the beta formula

If debt is risky, $\beta_D \neq 0$ and (1.12) keeps its last term. Amaro de Matos (§2.1.2, p. 43) writes the levered equity beta as $\beta_s = \beta_s^u(1 + \frac{B}{S}) - \beta_d \frac{B}{S}$, which is (1.12) with $S = E$, $B = D$, $\beta_s^u = \beta_A$. Setting $\beta_D = 0$ when debt is risky **overstates** the asset beta: the equity beta absorbs part of the risk that the debt in fact bears.

Links

- Next: which cash flows to discount → L02-02-cash-flows-npv-irr. The identity (1.10) at work with state prices → L02-05-entrepreneur-example §5; MM II is (1.12) with r_A fixed → L02-07-mm2-wacc-cost-of-debt.
- Topic 1 supplies $A = D + E$ in market value → L01; taxes change the unlevering formula → L03.
- Rules: rules/02 — valuation.

- Block map: Map/lectures-index (Topic 2A, blocks 2A.1–2A.2).
- **Practice.** lista-r2 #2 (Berk Ch. 12 Ex. 24, p. 377), #5 (Higgins Ch. 8 Ex. 7, p. 328), #6 (Grinblatt–Titman Ch. 11 Ex. 11.10–11.13, p. 416), #9 (Berk Ch. 13 Ex. 20, p. 405); PS2 Ex. 4(a) (PS2, pp. 2–3); quiz Topic 2A, block `val`.

2A.3–2A.4 · Which cash flows, the NPV rule, and why the IRR misleads

> **Claim.** Discount the **unlevered free cash flow** — what the assets throw off for all investors, computed as if all-equity — on an **incremental** basis. Accept projects with positive NPV: in a frictionless all-equity firm, the NPV is exactly the gain of the current shareholders, **whichever way the project is financed**. The IRR agrees with the NPV only for conventional, independent projects; otherwise it can have several values, none, or the wrong sign.

Prev: L02-01-present-value-capm-betas · Next: L02-03-debt-vs-equity-claims · Map: L02-00-map

Notation.

Symbol	Meaning	Range
FCF_t	unlevered free cash flow at t	$\mathbb{R}$
EBIT_t, τ	earnings before interest and taxes; effective tax rate	$\tau \in [0, 1)$
Dep_t, CAPEX_t	depreciation; capital expenditure	≥ 0
NWC_t	net working capital = receivables + inventories − payables	$\mathbb{R}$
g	perpetual growth rate of FCF after the horizon, $g < r$	—
TV_T, SV, PPE	terminal value at T; salvage value; book value of the assets sold	—
c_A, PV_A, NPV_A	cost at $t = 0$; value of the cash flows; $PV_A - c_A$	—
IRR	rate solving $NPV = 0$	> -1
α	share of the firm sold to new shareholders	$[0, 1]$
F	face value of debt (here $F = 16.5$)	≥ 0

---

1 · C — The free cash flow (slides 24–25)

Investors care about the cash they can extract. The cash available to **all** investors is

$$\text{FCF}_t = \text{EBIT}_t(1 - \tau) + \text{Dep}_t - \text{CAPEX}_t - \Delta\text{NWC}_t \tag{2.1}$$

with NWC_t = accounts receivable + inventories − accounts payable. Projections of FCF_t stand in for $E[\tilde{d}_t]$ in (1.1).

> **What is left out, and why.** (2.1) taxes EBIT **as if the firm had no debt**: subsidies created by financing — the interest tax shield — are not in the cash flow. They are accounted for **in the discount rate** (slide 25; the after-tax WACC of L03). Adding them to the cash flow as well would count them twice.

U-layer: Berk writes the same cash flow as FCF = (Revenues−Costs)(1−τ)−CAPEX−ΔNWC+τ Dep (Eq. 9.7, p. 259): depreciation matters only through the **depreciation tax shield** τ Dep. His example: at $\tau = 40\%$, a depreciation of 200,000 saves $0.40 \times 200{,}000 = 80{,}000$ of tax, raising FCF from 600,000 to 680,000.

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2 · C — Incremental, opportunity, sunk, inflation, terminal value (slides 26–27)

1. **Incremental.** Cash flow *with* the project minus cash flow *without* it.
2. **Opportunity costs are in:** a resource the firm already owns is charged at its value in its best alternative use.
3. **Sunk costs are out:** costs the firm is already liable for whatever it decides (Ross §10.2, p. 320: *"tal custo é claramente irrelevante para a decisão em questão"*; Higgins p. 263 drops 2 million of already-incurred preliminary engineering by the *with–without principle*).
4. **Inflation consistently:** nominal flows at nominal rates, real flows at real rates.
5. **Horizon.** Project 5–10 years, then a terminal value: liquidation, or growth at a constant rate in perpetuity.
6. **Salvage value** SV = liquidation price − liquidation costs, **taxed on** $SV - \text{PPE}$ (the gain over book value):
$$\text{after-tax salvage} = SV - \tau\,(SV - \text{PPE}) \tag{2.2}$$
7. **NWC is recouped** at the project's end: the last year carries $+\text{NWC}_{T-1}$.

The growing-perpetuity terminal value, derived. If $\text{FCF}_{T+k} = \text{FCF}_{T+1}(1+g)^{k-1}$ for $k \geq 1$, then at date T

$$\text{TV}_T = \sum_{k=1}^{\infty} \frac{\text{FCF}_{T+1}(1+g)^{k-1}}{(1+r)^k} = \frac{\text{FCF}_{T+1}}{1+r} \sum_{k=0}^{\infty} \Big(\frac{1+g}{1+r}\Big)^k$$

The geometric series converges iff $(1+g)/(1+r) < 1$, i.e. $g < r$, to $1/(1 - \frac{1+g}{1+r}) = \frac{1+r}{r-g}$:

$$\text{TV}_T = \frac{\text{FCF}_{T+1}}{1+r} \cdot \frac{1+r}{r-g} = \frac{\text{FCF}_{T+1}}{r-g} \tag{2.3}$$

and it enters (1.3) discounted by $(1+r)^T$.

---

3 · C — The NPV rule (slides 28–30)

$$NPV_A = PV_A - c_A \tag{2.4}$$

Rule: undertake the project iff $NPV_A > 0$.

- PV_A is the value of the cash flows **inside** the firm; c_A is what they cost **outside** it. A positive NPV is an *"arbitrage" across the boundary of the firm.*
- It can survive in equilibrium because, unlike a financial arbitrage, it **is not scalable**.
- It enriches, in expectation, the **current claimants** — equity and debt together. How they split it can distort decisions (debt overhang, later in the course).

- **Acquisitions (slide 30).** Buying target T at seller S's price c_T is positive-NPV iff $PV_T > c_T$, where PV_T is T's value *inside* acquirer P: synergies, scale/scope, better management, or market power can make it exceed c_T.

Berk states the same rule as *"take the alternative with the highest NPV. Choosing this alternative is equivalent to receiving its NPV in cash today"* (p. 213); Ross §9.1 (p. 282) and Higgins (p. 246) build it from the time value of money.

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4 · C — Worked example: NPV = shareholders' gain, three financings (slides 31–36)

Setting. Risk-neutral agents (or a project with $\beta = 0$), so every discount rate is $r_f = 10\%$; no taxes; all-equity firm; symmetric information. Assets in place are worth 100 (boom) or 10 (slump), probability 0.5 each. A safe project costs 15 at $t = 0$ and pays 22 in both states.

$$NPV = -15 + \frac{22}{1.10} = -15 + 20 = 5 \tag{2.5}$$

(a) Insiders pay the 15.

$$E_{\text{without}} = \frac{0.5 \times 100 + 0.5 \times 10}{1.10} = \frac{55}{1.10} = 50$$
$$E_{\text{with}} = \frac{0.5 \times (100 + 22) + 0.5 \times (10 + 22)}{1.10} = \frac{61 + 16}{1.10} = \frac{77}{1.10} = 70$$
$$\text{gain} = (70 - 50) - 15 = 5 = NPV$$

(b) New equity. Competitive investors pay 15 for a share α and break even:

$$\alpha \cdot \frac{0.5 \times (100 + 22) + 0.5 \times (10 + 22)}{1.10} = 15$$
$$\alpha \cdot 70 = 15 \implies \alpha = \frac{15}{70}$$

Old shareholders keep $1 - \alpha = 55/70$:

$$\frac{55}{70} \times 70 = 55, \qquad \text{gain} = 55 - 50 = 5 = NPV$$

(c) Safe debt. Lenders pay 15 for a promise F that is paid in both states (the slump payoff $10 + 22 = 32 > F$):

$$\frac{F}{1.10} = 15 \implies F = 16.5$$
$$E = \frac{0.5 \times (100 + 22 - 16.5) + 0.5 \times (10 + 22 - 16.5)}{1.10} = \frac{0.5 \times 105.5 + 0.5 \times 15.5}{1.10} = \frac{60.5}{1.10} = 55$$
$$\text{gain} = 55 - 50 = 5 = NPV$$

Which assumptions deliver "NPV = shareholders' gain"? Competitive outside investors (they take zero NPV, so the whole NPV stays inside), symmetric information (they price the new claim correctly), and **no risky debt outstanding** (no old creditor captures part of the gain). Slide 31 flags both failure cases: **leverage (risky debt)** and **asymmetric information**. Without frictions, the NPV rule *is* shareholder-value maximisation (slide 36).

---

5 · C — The IRR and its pitfalls (slides 37–38)

$$0 = -c_A + \sum_{t=1}^{T} \frac{E[\tilde{d}_t]}{(1+\text{IRR})^t} \tag{2.6}$$

IRR rule: accept iff IRR > discount rate. (2.6) is a polynomial of degree T in $x = 1/(1+\text{IRR})$; the pitfalls are properties of its roots.

(i) Multiple roots — Ross' mine (pp. 297–298). Cash flows −60, +155, −100:

$$NPV(r) = -60 + \frac{155}{1+r} - \frac{100}{(1+r)^2}$$

Multiply by $-(1+r)^2$ and set $u = 1+r$: $60u^2 - 155u + 100 = 0$, so

$$u = \frac{155 \pm \sqrt{155^2 - 4\cdot 60\cdot 100}}{2\cdot 60} = \frac{155 \pm \sqrt{24025 - 24000}}{120} = \frac{155 \pm 5}{120}$$

$$u \in \{1.25,\ 1.3333\} \implies \text{IRR} \in \{25\%,\ 33\tfrac{1}{3}\%\}$$

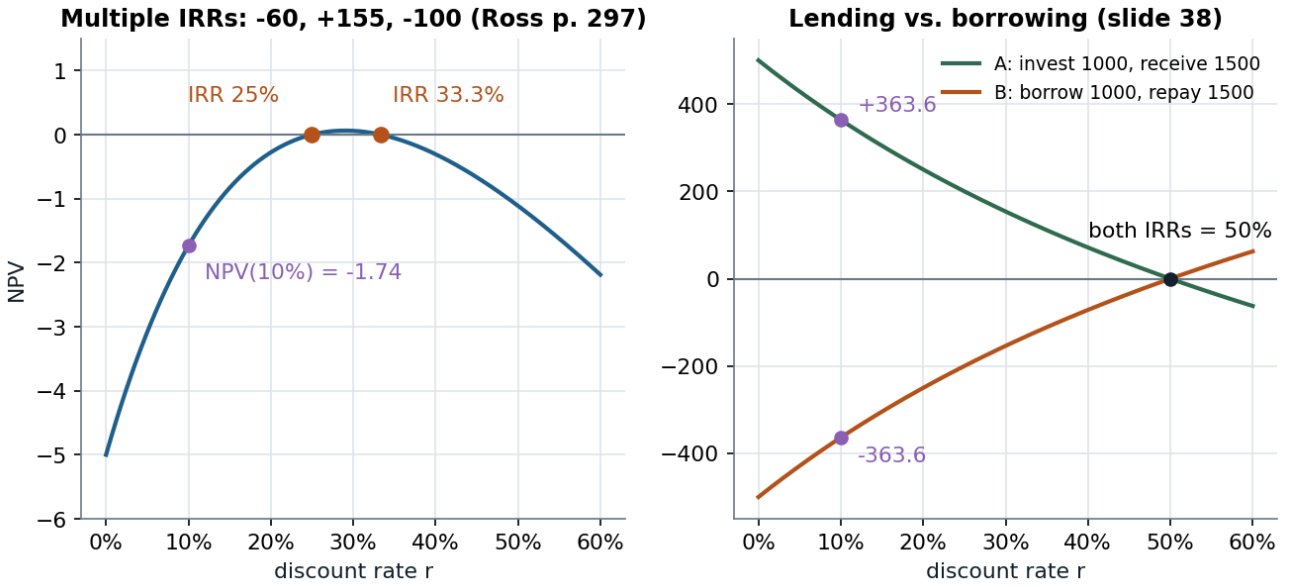
At $r = 10\%$: $NPV = -60 + 140.91 - 82.64 = -1.74$ (Ross' table). **Both IRRs exceed 10%, yet the NPV is negative** — the IRR rule says accept, the NPV rule says reject.

(ii) No root. *Illustration chosen here:* +100, −300, +250. With $x = 1/(1+r)$, $NPV = 100 - 300x + 250x^2$ has discriminant $300^2 - 4\cdot 250\cdot 100 = 90{,}000 - 100{,}000 < 0$: the NPV is positive at every rate and no IRR exists (Higgins p. 258 names the case).

(iii) Lending vs. borrowing (slide 38).

Project	$t=0$	$t=1$	IRR	NPV at 10%
A	−1000	+1500	50%	$-1000 + 1500/1.1 = +363.6$
B	+1000	−1500	50%	$1000 - 1500/1.1 = -363.6$

Same IRR, opposite NPVs: A *lends* at 50% (good), B *borrows* at 50% (bad).



Left: two IRRs, negative NPV at the 10% cost of capital (Ross pp. 297–298). Right: one IRR for a loan and a borrowing (slide 38). Eq. (2.6).

Read the left panel at the cost of capital, not at the roots: the NPV is positive only between 25% and 33⅓%. On the right, the sign of the slope tells lending (NPV falls with r) from borrowing (NPV rises with r); the IRR alone cannot.

(iv) Scale, mutually exclusive projects (slide 38).

Project	$t=0$	$t=1$	IRR	NPV at 10%
A	-10	14.3	$14.3/10-1=43\%$	$-10+14.3/1.1=3$
B	-1000	1210	$1210/1000-1=21\%$	$-1000+1210/1.1=100$

The IRR ranks A first; the NPV, which measures the wealth created, ranks B first. Higgins (p. 258) calls this *"a second, more serious problem"*; Berk works multiple IRRs on Evan's deal (IRRs 5.79% and 13.80%, pp. 221–223).

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6 · G — Why the NPV rule is preference-free: Fisher separation

Amaro de Matos (§1.1.5, pp. 11–12) adds an investment schedule with marginal return $\phi(I)$ to a consumer with access to a capital market at rate r. The first-order condition in the investment level does not involve the utility function:

$$\epsilon^* = \begin{cases} 0 & r \geq \phi(0) \\ \phi^{-1}(r) & \phi(0) \geq r \end{cases}$$

Proposition 1.1.1 (Fisher separation, p. 12): the optimal production decision is fixed by an objective market criterion (invest until the marginal return equals r), independent of preferences. §1.1.6 (p. 14) shows the investment raises current wealth by $\Delta/(1+r)$, with $\Delta = \int_0^{\phi^{-1}(r)} [\phi(I) - r]\, dI$ — **the**

NPV. Everyone who owns the firm agrees to maximise it, whatever their time preferences. This is the foundation slide 36 states in words.

Links

- Prev: the discount rate → L02-01-present-value-capm-betas. The financing-irrelevance in §4 is a special case of MM I → L02-06-mm1-arbitrage-put-call.
- FCF and NWC definitions come from Topic 1 → L01; the tax shield left out of (2.1) returns in L03.
- Rules: rules/02 — valuation.
- Python: `npv`, `irr_roots` in L02-10-python.
- **Practice.** PS1 Ex. 2 (incremental FCF, sunk fee, NWC; PS1 p. 1) and Ex. 3 (replacement decision, tax on sale vs. book value; pp. 1–2). lista-r2 #1 (Berk Ch. 8 Ex. 2, p. 241), #3 (Berk Ch. 9 Ex. 18, p. 276), #7 (Berk Ch. 8 Ex. 26, p. 244), #8 (Berk Ch. 8 Ex. 17, p. 243), #10 (Higgins Ch. 8 Ex. 16, p. 330), #11 (Berk Ch. 9 Ex. 25, p. 277), #12 (Higgins Ch. 9 Ex. 11, p. 382), #13 (Ross Cap. 10 Ex. 31, p. 356), #15 (Amaro 1.2.6 #4, p. 28); roadmap roteiro-r2.

2A.5 · Debt and equity: two claims on one payoff, and equity as a call

Claim. Capital structure is the composition of the claims on the firm's assets. Debt is a fixed, senior, finite-maturity promise without votes; equity is the residual, perpetual, voting claim. Under limited liability their payoffs are $\min(\tilde{A}, F)$ and $\max(\tilde{A} - F, 0)$: **they always add up to** $\tilde{A}$, debt is concave in the assets and equity is a **call option** on them with strike F.

Prev: L02-02-cash-flows-npv-irr · Next: L02-04-complete-markets-state-prices · Map: L02-00-map

Notation.

Symbol	Meaning	Range
$\tilde{A}$	value of the firm's assets at maturity $t = 1$	≥ 0
F	face value of a one-period zero-coupon debt contract	≥ 0
$\tilde{C}_D, \tilde{C}_E$	payoffs to debt and to equity at $t = 1$	≥ 0
$C(F)$, $P(F)$	price today of a call / put on the assets with strike F	≥ 0
r_f	risk-free rate	—

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1 · C — Four differences, and the contractual fringe (slides 39–42)

	Debt	Equity
(i) Payoff	fixed promised payments at fixed dates	variable: residual claimant
(ii) Seniority	senior: paid first	paid only after debt promises are met
(iii) Maturity	always fixed	perpetual
(iii′) Legal rights	no control rights	voting rights

(The slides number the fourth item "(iii)" twice.) Around the core:

- **Covenants** protect lenders without votes: limits on dividends, repurchases, new (especially senior or secured) debt; clauses transferring control if maximum leverage, minimum net worth (book assets − book liabilities) or minimum liquidity are breached. A breach other than missing principal or interest is a **technical default** (slide 41).
- **Default** lets the lender seize the firm's assets and take control, leaving equity with nothing. **Limited liability**: only the firm's assets can be seized, never the owners' personal assets. Lenders can also report the default to a **credit bureau** (slide 42).

G-layer: Tirole's Figure 2.3 (p. 77) ranks the claims by priority — secured debt, ordinary debt, subordinated debt, preferred stock, common stock — and adds hybrids such as convertible debt, which he reads as protection for lenders against risk-taking (p. 77).

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2 · C — The payoffs (slides 43–45)

The firm finances itself at $t = 0$ with the owner's money and a one-period debt promising F at $t = 1$; $\tilde{A}$ may be above or below F, so **the debt is risky**.

- $\tilde{A} < F$: default; senior creditors take everything, $\tilde{A}$; equity gets 0.
- $\tilde{A} \geq F$: creditors get F; equity gets $\tilde{A} - F$.

$$\tilde{C}_D = \min(\tilde{A}, F) \tag{3.1}$$

$$\tilde{C}_E = \max(\tilde{A} - F, 0) \tag{3.2}$$

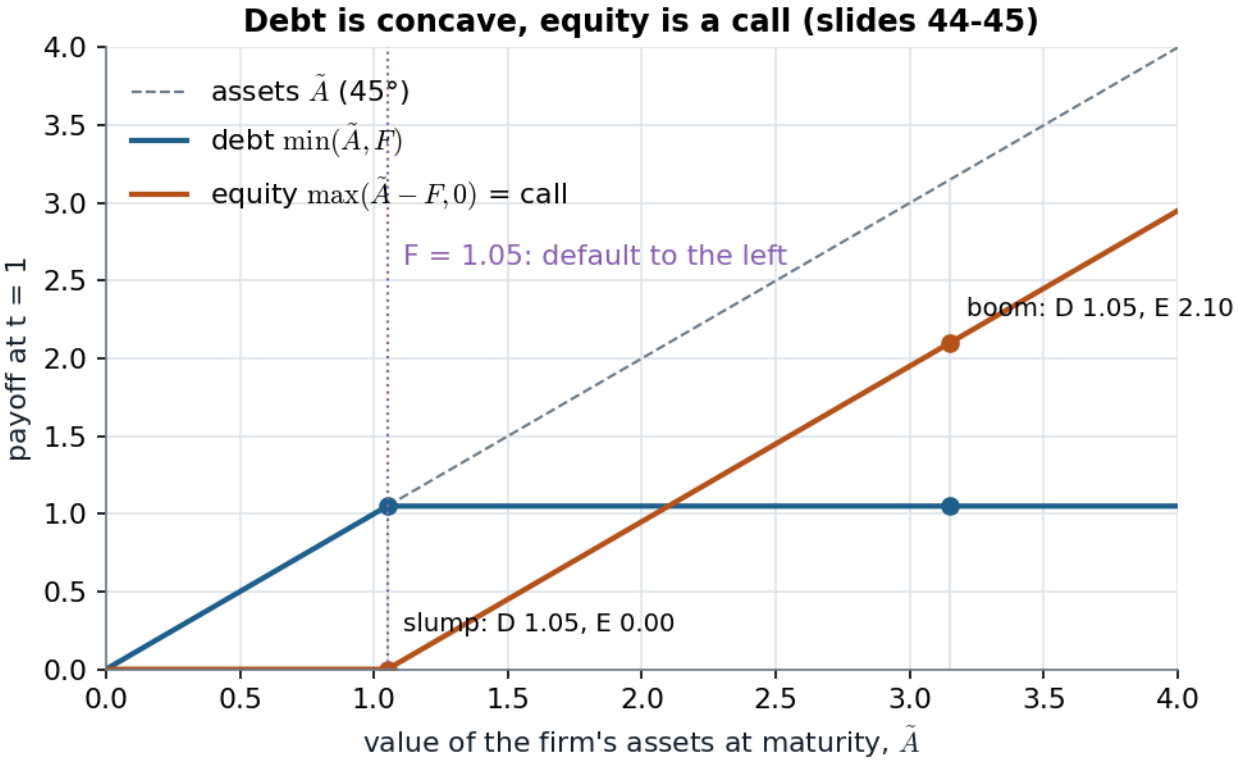
They add up. If $\tilde{A} \geq F$: $F + (\tilde{A} - F) = \tilde{A}$. If $\tilde{A} < F$: $\tilde{A} + 0 = \tilde{A}$. So

$$\tilde{C}_D + \tilde{C}_E = \tilde{A} \quad \text{in every state.} \tag{3.3}$$

(3.2) is, term for term, the payoff of a **call on $\tilde{A}$ with strike F**, so equity's value is $C(F)$. (3.1) can be rewritten (proved in L02-06-mm1-arbitrage-put-call §4):

$$\min(\tilde{A}, F) = F - \max(F - \tilde{A}, 0) \tag{3.4}$$

a riskless bond of face F **minus a put** on the assets with strike F.



Debt (3.1) is concave, equity (3.2) is convex and their sum is the 45° line (3.3); dots are the entrepreneur's project financed with debt of face 1.05 (slides 45, 58).

The kink at F is the whole story: left of it creditors bear the losses one for one, right of it shareholders take every extra unit. In the entrepreneur's economy, face 1.05 sits exactly at the slump payoff, so the debt never defaults and equity is worth something only in the boom.

Worked (slides 55, 58). Assets pay 1.05 (slump) or 3.15 (boom); $F = 1.05$. From (3.1)–(3.2):

	Slump	Boom
$\tilde{A}$	1.05	3.15
Debt $\min(\tilde{A}, 1.05)$	1.05	1.05
Equity $\max(\tilde{A} - 1.05, 0)$	0	2.10
Sum (3.3)	1.05	3.15

With $F = 2.1$ instead, debt pays $\min(1.05, 2.1) = 1.05$ and $\min(3.15, 2.1) = 2.1$ and **defaults in the slump**; equity pays 0 and 1.05.

---

3 · U — Why the call shape creates a conflict (slide 45, forward)

A convex payoff gains from spreading outcomes; a concave one loses. *Illustration chosen here:* keep $F = 1.05$ and replace the asset outcomes $(1.05,\ 3.15)$ by $(0,\ 4.2)$, same mean $\frac{1}{2}(1.05{+}3.15) = \frac{1}{2}(0{+}4.2) = 2.1$, states equally likely.

	Mean before	Mean after
Equity $\max(\tilde{A} - 1.05, 0)$	$\frac{1}{2}(0) + \frac{1}{2}(2.1) = 1.05$	$\frac{1}{2}(0) + \frac{1}{2}(3.15) = 1.575$
Debt $\min(\tilde{A}, 1.05)$	$\frac{1}{2}(1.05) + \frac{1}{2}(1.05) = 1.05$	$\frac{1}{2}(0) + \frac{1}{2}(1.05) = 0.525$

Total expected payoff is unchanged (2.1); **0.525 moves from creditors to shareholders.** Tirole (p. 77): *"(diversified) equityholders like (mean-preserving) increases in risk while debtholders dislike such increases in risk."* That is why covenants exist; the agency topics take it up.

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4 · G — The first irrelevance proof, in one line (Tirole p. 78)

With risk-neutral investors and repayment D on income R, $V_E{+}V_D = E[\max(0, R{-}D)]{+}E[\min(R, D)] = E[R]$, independent of D: (3.3) under the expectation. Tirole's footnotes 8–9 add that risk neutrality is not needed — with risk-averse investors one prices each state with *state-contingent prices* and applies the same equality. That is the route L02-04-complete-markets-state-prices and L02-05-entrepreneur-example take; Amaro de Matos does it in general (p. 41: $B_j(T) = \min\{F, V_j(T)\}$, eq. 2.2: $S_j(T) = \max[0, V_j(T) - F]$, so $V_j(T) = S_j(T) + B_j(T)$).

Links

- The put–call rewriting (3.4) and MM I with risky debt → L02-06-mm1-arbitrage-put-call §4.
- Pricing these payoffs state by state → L02-04-complete-markets-state-prices.
- Risk-shifting as a cost of distress → L03.

- Rules: rules/03 — MM benchmark.
- Interactive: The State-Price Balance Sheet (`companions/companion-mm-state-prices.html`) — move F across the slump payoff and watch the debt turn risky.
- **Practice.** PS2 Ex. 1 (calls and puts on the project's assets; PS2 p. 1); lista-r3 #11 (Grinblatt–Titman Ex. 14.2–14.3, pp. 526–527).

2A.6 · The frictionless benchmark, contingent claims and state prices

Claim. The benchmark's assumptions are chosen so that **a security is nothing but its payoff vector.** In a complete market every payoff vector can be built from traded securities, and no arbitrage then forces one **positive price per state**. Any asset's price is its payoffs times those state prices — a linear rule. Every later MM result is this linearity applied to debt and equity.

Prev: L02-03-debt-vs-equity-claims · Next: L02-05-entrepreneur-example · Map: L02-00-map

Notation.

Symbol	Meaning	Range
$s \in S = \{s_1, \dots, s_N\}$	states of the world at $t = 1$	N finite
cc_s	contingent claim: pays 1 in state s, 0 otherwise	—
p_s	price at $t = 0$ of cc_s (state price)	> 0
x, y	units of stock and bond in a replicating portfolio	$\mathbb{R}$ (short allowed)
S_0, B_0	prices of the stock and the bond	> 0
$\pi_s = p_s / \sum_k p_k$	risk-neutral probability of state s	$(0, 1)$
q_s	true probability of state s	$(0, 1)$

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1 · C — The strong assumptions (slides 46–47, repeated on 82)

In a frictionless benchmark the capital structure is irrelevant (MM). The assumptions:

1. **Competitive and complete markets**: price-takers; the cash flow of any (new) security can be replicated by existing ones.
2. Individuals and firms trade **at the same prices** (e.g. borrow at the same rate).
3. **No taxes, bankruptcy or other transaction costs.**
4. **Financing neither affects the assets' cash flows nor conveys information.**
5. **Fixed (exogenous) operating cash flow** — investment is given.
6. **Symmetric information.**

What each one buys. (1) makes every claim priceable by replication; (2) makes homemade leverage a perfect substitute for corporate leverage; (3) keeps the total payoff the same for every split; (4)–(5) keep the "pie" fixed while it is sliced; (6) lets outsiders price new claims correctly. Relaxing (3) gives L03; relaxing (4)–(6) gives the agency and asymmetric-information topics.

---

2 · C — States, contingent claims, completeness, no arbitrage (slides 48–51)

A **state** is a scenario for tomorrow ($\{\text{sun}, \text{rain}\}$; GDP growth $\{-3\%, 0\%, +3\%\}$). A **contingent claim** cc_s pays 1 in state s and 0 otherwise; with three states the three claims are the rows of the identity matrix (slide 49).

- **Complete market:** every cc_s can be **synthesised** by buying and short-selling traded securities — the claims need not trade themselves (slide 50). Amaro de Matos (p. 21): a unique replicating

portfolio exists for every payoff iff the number of securities with linearly independent payoffs equals the number of states, $N = K$.

- **Law of one price:** assets or portfolios with the same payoff trade at the same price.
- **Arbitrage** (slide 51): a *free lunch* — positive cash today, no obligation tomorrow; or a *money machine* — zero cost today, a payoff ≥ 0 in every state and > 0 with positive probability. It cannot survive if a single rational investor prefers more to less.

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3 · C — Worked example: state prices from a stock and a bond (slides 52–54)

	Price $t = 0$	Slump	Boom
Stock	2	1.9	2.4
Bond	1	1.05	1.05

A portfolio of x stocks and y bonds pays $1.9x + 1.05y$ (slump) and $2.4x + 1.05y$ (boom) and costs $2x + y$.

cc_1 (boom claim): pay $(0, 1)$.

$$1.9x + 1.05y = 0 \tag{4.1}$$

$$2.4x + 1.05y = 1 \tag{4.2}$$

Subtract (4.1) from (4.2): $0.5x = 1$, so $x = 2$. Substitute into (4.1): $1.9(2) + 1.05y = 0$, so $y = -3.8/1.05$ (short 3.8/1.05 bonds). By the law of one price,

$$p_1 = 2x + y = 2(2) + \left(-\frac{3.8}{1.05}\right) = 4 - 3.6190 = 0.381 \tag{4.3}$$

cc_2 (slump claim): pay $(1, 0)$.

$$1.9x + 1.05y = 1, \qquad 2.4x + 1.05y = 0$$

Subtracting: $-0.5x = 1$, so $x = -2$; then $2.4(-2) + 1.05y = 0$, so $y = 4.8/1.05$:

$$p_2 = 2(-2) + \frac{4.8}{1.05} = -4 + 4.5714 = 0.571 \tag{4.4}$$

Check: the bond. A riskless payoff of 1.05 in both states is $1.05\,cc_1 + 1.05\,cc_2$: $1.05(0.380952 + 0.571429) = 1.05 \times 0.952381 = 1.000$. So $p_1 + p_2 = 1/1.05$, the price of a sure unit, and $r_f = 5\%$.

Any asset (slide 54). An asset paying x in the slump and y in the boom is y units of cc_1 plus x units of cc_2:

$$\text{price} = 0.381\,y + 0.571\,x \tag{4.5}$$

The price operator is linear in payoffs. (4.5) is the whole content of complete markets plus no arbitrage; MM I is (4.5) applied to $\min(\tilde{A}, F) + \max(\tilde{A} - F, 0) = \tilde{A}$.

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4 · U — Why state prices are not probabilities

The two states may be equally likely, yet $p_2 > p_1$: a unit in the slump costs more. Slide 67 gives the reason in the next example: risk-averse agents' marginal utility is higher when they are poorer, so they pay more for consumption in bad times.

Normalising the state prices gives **risk-neutral probabilities** (Amaro de Matos eq. 1.13, p. 24):

$$\pi_s = \frac{p_s}{\sum_k p_k}, \qquad \text{price} = \frac{1}{1+r_f}\sum_s \pi_s \,\text{payoff}_s \tag{4.6}$$

Here $p_1 = 0.4/1.05$ and $p_2 = 0.6/1.05$ exactly (from (4.3): $4-3.8/1.05 = (4.2-3.8)/1.05$), so $\pi_{\text{boom}} = 0.4$ and $\pi_{\text{slump}} = 0.6$: the bad state is over-weighted relative to any "true" 50/50. Amaro de Matos (p. 25) stresses that π *"does not necessarily correspond to the vector of real probabilities"*; with the true ones, one discounts at a risk-adjusted rate instead, or weights by the **state-price deflator** $\delta_j = p_j/q_j$, the ratio of marginal utilities (p. 25).

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5 · G — The theorems behind the example

- **Proposition 1.2.1 (Amaro de Matos, p. 22):** absence of arbitrage is equivalent to the existence of a strictly positive state-price vector with $V = \Phi\psi$ (Farkas' lemma).
- **Uniqueness** needs completeness: then $\psi = \Phi^{-1}V$ (p. 23). With $N < K$ several state-price vectors price the traded assets, and a new payoff can have a range of no-arbitrage values (p. 24).
- **§1.2.3 (p. 23):** with complete markets and no arbitrage, every *financial* investment has zero NPV — positive NPV can only come from real projects whose payoffs are not yet spanned at their cost, the "arbitrage across the firm's boundary" of L02-02-cash-flows-npv-irr §3.

Links

- The same machinery on the entrepreneur's firm → L02-05-entrepreneur-example; on any capital structure → L02-06-mm1-arbitrage-put-call.
- Rules: rules/03 — MM benchmark.
- Interactive: The State-Price Balance Sheet (`companions/companion-mm-state-prices.html`) — set p_b, p_s and read $r_f = 1/(p_b + p_s) - 1$ and every value.
- Python: `state_prices` (solves (4.1)–(4.2) for any two securities) in L02-10-python.
- **Practice.** PS2 Ex. 1(a) (state prices in a complete two-state economy; PS2 p. 1); lista-r2 #14 (Amaro 1.2.6 #5, p. 28) and #15 (Amaro 1.2.6 #4, p. 28).

2A.7 · The penniless entrepreneur: one firm, every financing, the same value

> **Claim.** Price the project with state prices recovered from a stock and a bond. The firm is worth $A = 1.75$ against a cost of 1, whether it is financed by debt, equity or any mix: outside investors break even, and the entrepreneur keeps the whole NPV of 0.75. The return on assets is fixed at 20%. **Leverage only moves risk between the claims**: the expected return on equity rises with D/E in exactly the way that keeps A constant. A table that holds r_E fixed while borrowing more manufactures value that is not there.

Prev: L02-04-complete-markets-state-prices · Next: L02-06-mm1-arbitrage-put-call · Map: L02-00-map

Notation.

Symbol	Meaning	Value / range
X_s, X_b	project payoff in slump, boom	1.05, 3.15
q	probability of each state	0.5
p_1, p_2	price of the boom claim cc_1, slump claim cc_2	0.357, 0.595
x, y	units of stock and bond in a replicating portfolio	$\mathbb{R}$
α	share of the firm sold to new shareholders for 1	$[0, 1]$
D	amount raised by debt ($1 - D$ by equity)	$[0, 1]$
A, D, E	market values of assets, debt, equity	—
r_A, r_D, r_E	expected returns: expected payoff / value − 1	—

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1 · C — Setting, and the book-vs-market question (slides 55–57)

The entrepreneur has no money and must raise 1 at $t = 0$. Two equally likely states:

	Investment $t = 0$	Slump	Boom	Expected return on the investment
Project	1	1.05	3.15	$\frac{1}{2}(1.05 + 3.15)/1 - 1 = 110\%$
Stock	price 1	0	2.8	$\frac{1}{2}(0 + 2.8)/1 - 1 = 40\%$
Bond	price 1	1.05	1.05	5%

Investors trade the stock and the bond freely, short sales included. Options: issue debt (it can be riskless — 1.05 is paid in both states), sell a share α of equity for 1, or raise D by debt and $1 - D$ by equity. **Is $r_A = 110\%$?** No: 1 is the **book** value of the assets, the cost. r_A is defined on the **market** value, which we do not know yet.

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2 · C — The state prices (slides 60–67)

A portfolio of x stocks and y bonds pays $1.05y$ in the slump and $2.8x + 1.05y$ in the boom and costs $x \cdot 1 + y \cdot 1$ (in general $xS_0 + yB_0$; slide 64).

cc_1, pays $(0, 1)$ (slide 65):

$$1.05y = 0 \tag{5.1}$$

$$2.8x + 1.05y = 1 \tag{5.2}$$

(5.1) gives $y = 0$; substituting into (5.2): $2.8x + 1.05(0) = 1$, so $x = 1/2.8$.

$$p_1 = x + y = \frac{1}{2.8} + 0 = 0.357 \tag{5.3}$$

cc_2, pays $(1, 0)$ (slide 66):

$$1.05y = 1 \implies y = \frac{1}{1.05}$$

$$2.8x + 1.05\Big(\frac{1}{1.05}\Big) = 0 \implies x = -\frac{1}{2.8} \quad \text{(short the stock)}$$

$$p_2 = x + y = -\frac{1}{2.8} + \frac{1}{1.05} = -0.357143 + 0.952381 = 0.595 \tag{5.4}$$

Same probability, different price (slide 67): $p_2 > p_1$. Agents are risk-averse; a unit in the slump, when they are poorer, is worth more to them than a unit in the boom. $p_1 + p_2 = 1/1.05$ recovers $r_f = 5\%$.

---

3 · C — Debt financing (slides 58, 61, 68)

Borrow 1 at 5%, promise $F = 1.05$. Payoffs (debt is senior):

	Slump	Boom
Total	1.05	3.15
Debt	1.05	1.05
Entrepreneur	0	2.10

The entrepreneur's position pays $(0,\ 2.1) =$ **2.1 units of** cc_1; by the law of one price,

$$\text{entrepreneur} = 2.1\, p_1 = 2.1 \times 0.357143 = 0.75 \tag{5.5}$$

$$A = 3.15\, p_1 + 1.05\, p_2 = 3.15(0.357143) + 1.05(0.595238) = 1.125 + 0.625 = 1.75 \tag{5.6}$$

Lending is zero-NPV for competitive creditors, so the market value of the debt equals the cash they put in, 1. **Under debt:** $A = 1.75$**,** $E = 0.75$**,** $D = 1$, and $A = D + E$.

---

4 · C — Equity financing (slides 59, 62–63, 69)

New shareholders pay 1 for α; the entrepreneur keeps $1-\alpha$:

	Slump	Boom
Total	1.05	3.15
New shareholders	1.05α	3.15α
Entrepreneur	$1.05(1-\alpha)$	$3.15(1-\alpha)$

The new holders' claim is 3.15α units of cc_1 plus 1.05α units of cc_2. **Zero NPV for competitive investors** (slide 63):

$$1 = \alpha\,(3.15\,p_1 + 1.05\,p_2) \tag{5.7}$$

Substitute (5.6), $3.15p_1 + 1.05p_2 = 1.75$:

$$1 = \alpha\,(1.75) \implies \alpha = \frac{1}{1.75} = \frac{4}{7} \tag{5.8}$$

The entrepreneur's position is $(1-\alpha)3.15$ units of cc_1 plus $(1-\alpha)1.05$ of cc_2:

$$\tfrac{3}{7}\,(3.15\,p_1) + \tfrac{3}{7}\,(1.05\,p_2) = \tfrac{3}{7}\,(1.75) = 0.75 \tag{5.9}$$

New holders' claim is worth 1, total equity $0.75 + 1 = 1.75$. **Under equity:** $A = E = 1.75$**.**

> **MM in one line** (slide 70): $A = 1.75$ under both financings; investors take zero NPV; the entrepreneur keeps the NPV $1.75 - 1 = 0.75$ either way.

---

5 · C — The slide-81 exercise: replicate the positions directly

Slide 81 asks to price the entrepreneur's positions with the stock and the bond, skipping the contingent claims. **Under debt**, target $(0,\ 2.1)$:

$$1.05y = 0 \ \Rightarrow\ y = 0; \qquad 2.8x + 1.05(0) = 2.1 \ \Rightarrow\ x = 0.75$$

$$\text{cost} = 0.75(1) + 0(1) = 0.75\ \checkmark$$

Under equity, target $\frac{3}{7}(1.05,\ 3.15) = (0.45,\ 1.35)$:

$$1.05y = 0.45 \ \Rightarrow\ y = 0.428571$$

$$2.8x + 1.05(0.428571) = 1.35 \ \Rightarrow\ 2.8x = 1.35 - 0.45 = 0.90 \ \Rightarrow\ x = 0.321429$$

$$\text{cost} = 0.321429 + 0.428571 = 0.75\ \checkmark$$

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6 · C — Returns under each financing (slides 71–72)

Expected payoffs use $q = 0.5$; each return is expected payoff / market value − 1.

Debt financing:

	Value	Slump	Boom	Expected payoff	Expected return
Debt	1	1.05	1.05	1.05	$1.05/1 - 1 = 5\% = r_D$
Equity	0.75	0	2.1	1.05	$1.05/0.75 - 1 = 40\% = r_E$
Assets	1.75	1.05	3.15	2.10	$2.10/1.75 - 1 = 20\% = r_A$

Check (1.10): $\frac{E}{D+E} r_E + \frac{D}{D+E} r_D = \frac{0.75}{1.75}(0.40) + \frac{1}{1.75}(0.05) = 0.171429 + 0.028571 = 0.20 = r_A$ ✓.

Equity financing ($D = 0$): $E = A = 1.75$, payoffs $(1.05, 3.15)$, $r_E = r_A = 20\%$. Check: $\frac{1.75}{1.75}(0.20) + 0 = 0.20$ ✓.

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7 · C — Any mix $D \in [0, 1]$: r_A fixed, r_E rising (slides 73–77)

For every $D \leq 1$ the face value $1.05D \leq 1.05$ is paid in both states, so debt is riskless and $r_D = 5\%$. The assets' payoffs and value do not depend on D, so neither does $r_A = E[\tilde{C}_A]/A - 1 = 20\%$ (slide 73). Rearranging (1.10) (slide 74):

$$r_E = \frac{E + D}{E} r_A - \frac{D}{E} r_D \tag{5.10}$$

Substitute $r_A = 20\%$, $r_D = 5\%$:

$$r_E = \frac{E + D}{E}(20\%) - \frac{D}{E}(5\%) = 20\% + \frac{D}{E}(20\%) - \frac{D}{E}(5\%) = 20\% + \frac{D}{E}\, 15\% \tag{5.11}$$

Substitute $E = 1.75 - D$ (slide 75):

$$r_E = 20\% + \frac{D}{1.75 - D}\, 15\% \tag{5.12}$$

Increasing and convex in D (slide 76), **linear in** D/E (slide 77): at $D = 1$, $r_E = 20\% + \frac{1}{0.75} 15\% = 40\%$.

$D = 0.4$ **(slide 78).** Debt pays $0.4 \times 1.05 = 0.42$ in both states; equity pays $1.05 - 0.42 = 0.63$ and $3.15 - 0.42 = 2.73$.

$$E = 0.63\, p_2 + 2.73\, p_1 = 0.63(0.595238) + 2.73(0.357143) = 0.375 + 0.975 = 1.35$$

$$r_E = \frac{\frac{1}{2}(0.63 + 2.73)}{1.35} - 1 = \frac{1.68}{1.35} - 1 = 24.44\%$$

Check (5.12): $20\% + \frac{0.4}{1.35} 15\% = 20\% + 4.44\% = 24.44\%$ ✓.

	Value	Slump	Boom	Expected return
Debt	0.40	0.42	0.42	5%
Equity	1.35	0.63	2.73	24.4%
Assets	1.75	1.05	3.15	20%

The fallacy (slide 79). Since $r_D < r_E$, borrow 0.5 and buy back equity? If the expected returns stayed at 5% and 24.4%, equity paying $(0.525,\ 2.625)$ would be valued at

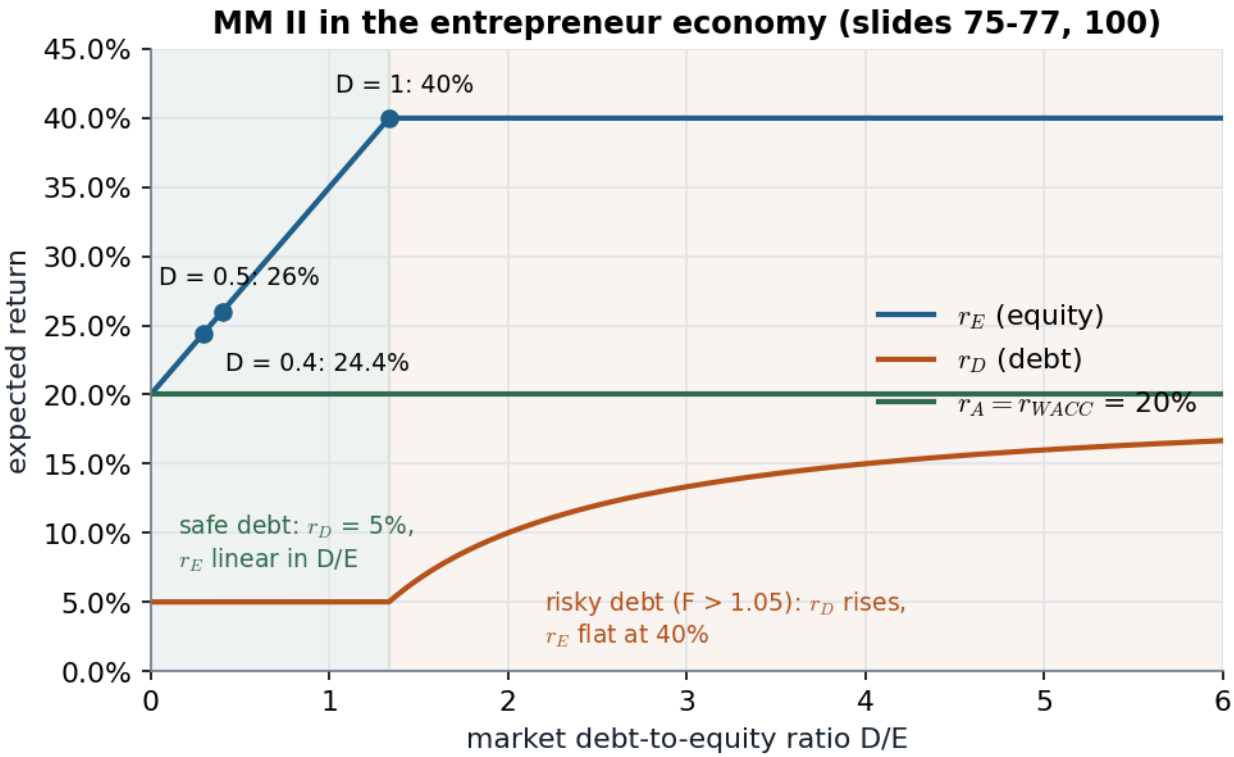
$$E^{\text{wrong}} = \frac{\frac{1}{2}(0.525 + 2.625)}{1.2444} = \frac{1.575}{1.2444} = 1.27, \qquad A^{\text{wrong}} = 0.50 + 1.27 = 1.77 > 1.75.$$

Correct values (slide 80) come from the state prices, which do not care about labels:

$$E = 0.525\,p_2 + 2.625\,p_1 = 0.525(0.595238) + 2.625(0.357143) = 0.3125 + 0.9375 = 1.25$$
$$r_E = \frac{1.575}{1.25} - 1 = 26\%, \qquad A = 0.5 + 1.25 = 1.75$$

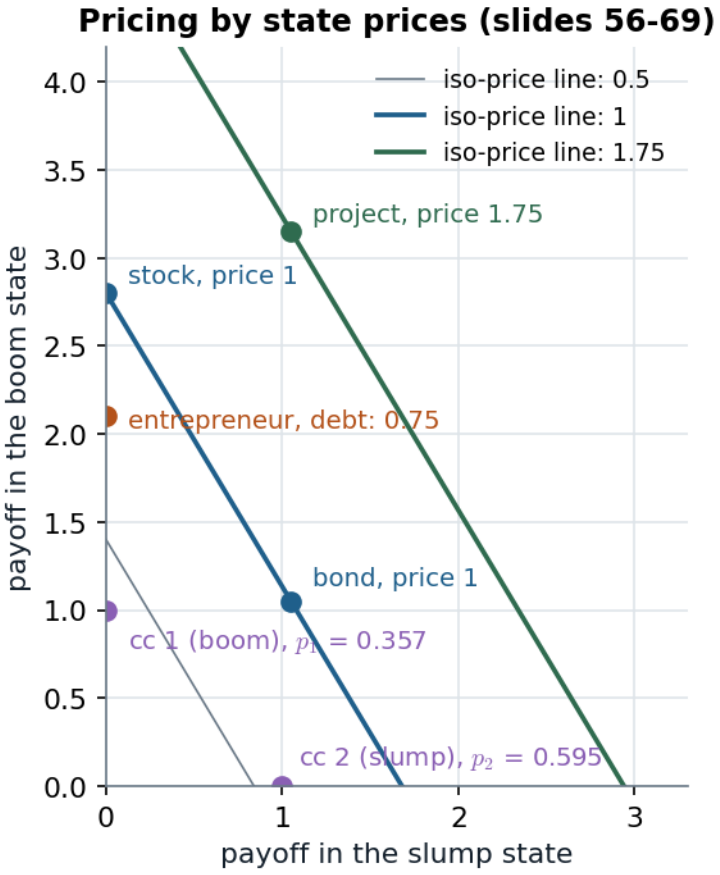
Check (5.12): $20\% + \frac{0.5}{1.25}15\% = 20\% + 6\% = 26\%$ ✓.

The expected return on equity must rise with leverage (slide 79). Holding it fixed is the single most common way to "prove" that debt creates value in a frictionless market.



The left, shaded part is slides 76–77 (safe debt, eq. 5.11); the right part extends the same economy past face 1.05, where debt defaults in the slump (eq. 7.6 in L02-07-mm2-wacc-cost-of-debt).

The dots are the three tables of this note: 24.4%, 26%, 40%. To the left of $D/E = 1.33$ the line is (5.11); past it, an extra unit of promised debt buys only boom payoffs, so equity becomes pure boom claims and its return freezes at $\frac{1}{2}/p_1 - 1 = 40\%$.



Price is linear in payoffs: p_2·slump + p_1·boom. Points on one line cost the same (slides 56, 60–69; eqs. 5.3–5.6).

The stock and the bond both cost 1, so the line through them **is** the price-1 line; its slope $-p_2/p_1 = -1.667$ is the market's rate of exchange between slump and boom payoffs. Any split of the project's point (1.05, 3.15) into two claims puts them on parallel lines whose prices add to 1.75.

> **Interactive companion.** The State-Price Balance Sheet (`companions/companion-mm-state-prices.html`) — load the slide presets ($D = 1$, 0.4, 0.5, all equity), then drag F: A never moves while D, E and r_E do.

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8 · U — The same story in dollars: Berk's coffee shop (pp. 465–469)

Investment 24,000; cash flow 34,500 in one year, risk-free rate 5% plus a 10% risk premium, so 15%. Unlevered value and NPV:

$$V^U = \frac{34{,}500}{1.15} = 30{,}000, \qquad NPV = 30{,}000 - 24{,}000 = 6{,}000$$

Borrow 15,000 at 5% (repay $15{,}000 \times 1.05 = 15{,}750$): levered equity is worth $30{,}000 - 15{,}000 = 15{,}000$ and expects $34{,}500 - 15{,}750 = 18{,}750$, a return of $18{,}750/15{,}000 - 1 = 25\%$, not 15%. Discounting it at

15% would give $18{,}750/1.15 = 16{,}304 > 15{,}000$ — the slide-79 fallacy in Berk's numbers (p. 467). With 6,000 of debt (repay 6,300), equity is 24,000 and expects $28{,}200/24{,}000 - 1 = 17.5\%$ (p. 469). Either way the founder keeps the 6,000 NPV.

Links

- The general proof by arbitrage → L02-06-mm1-arbitrage-put-call; MM II and the risky region of the figure → L02-07-mm2-wacc-cost-of-debt.
- The identity (1.10) used in §6–7 → L02-01-present-value-capm-betas §5.
- Topic 2B adds taxes to this firm → L03.
- Rules: rules/03 — MM benchmark.
- Python: `state_prices`, `claims`, `alpha_equity` with the slide tables as asserts → L02-10-python.
- **Practice.** PS2 Ex. 1(e) (raising 1, or D, by debt; PS2 p. 1) and PS2 Ex. 3(a–b) (all-equity value, face value of debt, equity value; p. 2); lista-r3 #1 (Berk Ch. 16 Ex. 16.1, p. 490).

2A.8 · MM I: the arbitrage proof, homemade leverage, and put–call parity

> **Claim.** In a frictionless economy the value of a firm does not depend on its capital structure (MM I). Proof by contradiction: if two firms with the same assets but different financing had different values, an investor could sell the dear one, rebuild it from the cheap one's securities — **undoing** or **creating** leverage on personal account — and pocket the difference. With risky debt the same fact is **put–call parity**: equity is a call, debt is a safe bond minus a put, and call + bond − put is the asset, whatever the strike.

Prev: L02-05-entrepreneur-example · Next: L02-07-mm2-wacc-cost-of-debt · Map: L02-00-map

Notation.

Symbol	Meaning	Range
X	random liquidating payoff of each firm at $t = 1$ (e.g. resale value)	> 0
E_1, D_1; $V_1 = D_1 + E_1$	firm 1: equity, **safe** debt, total value	> 0
E_2; $V_2 = E_2$	firm 2: all equity, total value	> 0
$R_f = 1 + r_f$	gross risk-free rate	> 0
α	fraction of a firm's securities traded by the arbitrageur	$(0, 1]$
K	face value of (possibly risky) debt	≥ 0
$C(K), P(K)$	prices of a call / put on X with strike K	≥ 0
$V(K)$	firm value when debt has face K	—

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1 · C — The statement (slides 82–85)

Assumptions: the six of L02-04-complete-markets-state-prices §1 (slide 82 repeats them).

> **MM I (slide 83).** *In a frictionless economy, total firm value is independent of the capital structure choice.*

Set-up (slides 84–85): firms 1 and 2 are identical except for financing; both pay X at $t = 1$. Firm 1 has equity E_1 and safe debt D_1; firm 2 has no debt. $V_1 = D_1 + E_1$, $V_2 = E_2$. Goal: $V_1 = V_2$. Complete markets let one go long or short each of E_1, D_1, V_1, E_2, V_2.

Payoffs at $t = 1$ (slide 86): firm 1's debtholders get $R_f D_1$, its shareholders $X - R_f D_1$; firm 2's shareholders get X.

---

2 · C — Safe debt: the two arbitrage steps (slides 86–89)

Step 1 — suppose $V_2 > V_1$. Sell the overpriced, buy the underpriced, matching payoffs:

Trade	Investment ($t=0$)	Payoff ($t=1$)
Short-sell α of firm 2's equity	$-\alpha V_2$	$-\alpha X$
Buy $\alpha V_2/V_1$ of firm 1's debt	$(\alpha V_2/V_1)D_1$	$(\alpha V_2/V_1)R_f D_1$
Buy $\alpha V_2/V_1$ of firm 1's equity	$(\alpha V_2/V_1)E_1$	$(\alpha V_2/V_1)(X - R_f D_1)$

Total investment:

$$-\alpha V_2 + \frac{\alpha V_2}{V_1}(D_1 + E_1) = -\alpha V_2 + \frac{\alpha V_2}{V_1}(V_1) = 0 \tag{6.1}$$

Total payoff — the $R_f D_1$ terms cancel:

$$-\alpha X + \frac{\alpha V_2}{V_1}R_f D_1 + \frac{\alpha V_2}{V_1}(X - R_f D_1) = -\alpha X + \frac{\alpha V_2}{V_1}X = \alpha X\Big(\frac{V_2}{V_1} - 1\Big) > 0 \tag{6.2}$$

A zero-cost portfolio paying a positive amount in every state: a **money machine**. Buying firm 1's debt **and** equity in proportion **undoes firm 1's leverage** (slide 88).

Step 2 — suppose $V_1 > V_2$.

Trade	Investment ($t=0$)	Payoff ($t=1$)
Buy $\alpha V_1/V_2$ of firm 2's equity	$(\alpha V_1/V_2)V_2 = \alpha V_1$	$(\alpha V_1/V_2)X$
Borrow αD_1 personally	$-\alpha D_1$	$-\alpha R_f D_1$
Short-sell α of firm 1's equity	$-\alpha E_1$	$-\alpha(X - R_f D_1)$

$$\text{investment} = \alpha V_1 - \alpha D_1 - \alpha E_1 = \alpha[V_1 - (E_1 + D_1)] = 0 \tag{6.3}$$

$$\text{payoff} = \frac{\alpha V_1}{V_2}X - \alpha R_f D_1 - \alpha X + \alpha R_f D_1 = \alpha X\Big(\frac{V_1}{V_2} - 1\Big) > 0 \tag{6.4}$$

The arbitrageur **levers up firm 2 on personal account** — homemade leverage (slide 89). Both steps fail only if $V_1 = V_2$. ■

> **Which assumption carries the proof?** The personal loan in Step 2 must cost the same R_f as the firm's: assumption 2 of the benchmark. With a higher personal borrowing rate, Step 2's payoff falls by $\alpha D_1(R_f^{\text{personal}} - R_f)$ and a band of values $V_1 > V_2$ survives (lista-r3 #15 works this case). Short sales (completeness) carry Step 1.

Numerical illustration (built here on the entrepreneur economy). Firm 1: $D_1 = 1$ (safe, repays $R_f D_1 = 1.05$), $E_1 = 0.75$, $V_1 = 1.75$; $X \in \{1.05,\ 3.15\}$. Suppose firm 2 traded at $V_2 = 1.80$. Step 1 with $\alpha = 0.1$: short 0.1 of firm 2 (+0.18 today); buy $0.1 \times 1.80/1.75 = 0.102857$ of firm 1's debt and equity, costing $0.102857 \times (1 + 0.75) = 0.18$. Net cost 0; payoff by (6.2) $0.1X(1.80/1.75 - 1) = 0.002857X$: 0.0030 in the slump, 0.0090 in the boom.

U-layer: Ross' Transmadruga (pp. 543, 545–546) and Berk's coffee shop (p. 469) make the same point for a single investor: whoever prefers the levered structure can borrow personally and buy the unlevered shares (*"alavancagem caseira"*); Berk: *"homemade leverage is a perfect substitute for the use of leverage by the firm"* when investors borrow at the firm's rate.

3 · C — Risky debt is a safe bond minus a put (slides 90–92)

With face K and $P(X<K)>0$, debt pays $\min\{K,X\}$. Case by case:

$$X \geq K: \quad K-\max\{K-X,0\} = K-0 = K = \min\{K,X\}$$
$$X < K: \quad K-\max\{K-X,0\} = K-(K-X) = X = \min\{K,X\}$$

$$\min\{K,X\} = \underbrace{K}_{\text{risk-free bond, face } K} - \underbrace{\max\{K-X,0\}}_{\text{put on the assets, strike } K} \tag{6.5}$$

By the law of one price:

$$D = \frac{K}{1+r_f} - P(K) \tag{6.6}$$

For safe debt every $X \geq K$, so $P(K)=0$ and $D = K/(1+r_f)$. $P(K)$ is the **price discount** of risky debt relative to a riskless bond of the same face (slide 92).

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4 · C — Equity is a call; parity gives MM I (slides 93–96)

Equity pays $\max\{X-K,0\}$, a call on X with strike K, so

$$E = C(K) \tag{6.7}$$

Add (6.6) and (6.7):

$$V(K) = \underbrace{C(K)}_{=E} + \underbrace{\frac{K}{1+r_f} - P(K)}_{=D} \tag{6.8}$$

Why (6.8) does not depend on K (slides 95–96). For any strike, the portfolio "buy the call, write the put, buy a bond of face K" pays:

	Investment	$X<K$	$X \geq K$
Buy call, strike K	$C(K)$	0	$X-K$
Write put, strike K	$-P(K)$	$-(K-X)$	0
Buy bond, face K	$K/(1+r_f)$	K	K
Total	$V(K)$	X	X

Every column sums to X. Two portfolios with strikes K_1, K_2 both pay X, so by no arbitrage $V(K_1) = V(K_2)$: **put–call parity is MM I with risky debt.**

Worked check in the entrepreneur economy ($p_1 = 1/2.8$ boom, $p_2 = 1/1.05 - 1/2.8$ slump, $X \in \{1.05, 3.15\}$, $1/(1+r_f) = p_1+p_2 = 0.952381$):

	$K = 1.05$ (safe)	$K = 2.1$ (risky)
$C(K) = p_1 \max(3.15 - K, 0) + p_2 \max(1.05 - K, 0)$	$0.357143 \times 2.1 = 0.75$	$0.357143 \times 1.05 = 0.375$
$K/(1 + r_f)$	$1.05 \times 0.952381 = 1.00$	$2.1 \times 0.952381 = 2.00$
$P(K) = p_1 \max(K - 3.15, 0) + p_2 \max(K - 1.05, 0)$	0	$0.595238 \times 1.05 = 0.625$
D by (6.6)	$1.00 - 0 = 1.00$	$2.00 - 0.625 = 1.375$
$V(K)$ by (6.8)	$0.75 + 1.00 - 0 = 1.75$	$0.375 + 2.00 - 0.625 = 1.75$

Interactive companion. The State-Price Balance Sheet (`companions/companion-mm-state-prices.html`) — the dashed box under the panels prints call + bond − put at the current F; drag F from 0 to 5 and the total stays at A.

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5 · G — The state-price proof in one line (Amaro de Matos pp. 41–42)

With payoffs $V_j(T)$ in state j and a strictly positive state-price vector ψ: $B_j(T) = \min\{F, V_j(T)\}$, $S_j(T) = \max[0, V_j(T) - F]$, and

$$V_l = S + B = \sum_j \psi_j [B_j(T) + S_j(T)] = \sum_j \psi_j V_j(T) = V_u$$

— **Proposition 2.1.1 (MM1):** *"The market value of any firm is independent of its capital structure,"* $V_l = V_u$ (eq. 2.3). Amaro de Matos notes it also means equity can be withdrawn for consumption and replaced by debt *"without any side effect on the total value of the firm"* (p. 42). Tirole (p. 78) proves it with risk-neutral expectations and adds the policy corollary: the financial structure is irrelevant, and so is payout policy.

Links

- The example that the proof generalises → L02-05-entrepreneur-example; the consequence for the cost of capital → L02-07-mm2-wacc-cost-of-debt.
- Payoff shapes (3.1)–(3.4) → L02-03-debt-vs-equity-claims.
- MM I with taxes, $V_L = V_U + \tau D$ → L03.
- Rules: rules/03 — MM benchmark.
- Python: `arbitrage_step1`, `put_call_value` with asserts → L02-10-python.
- **Practice.** PS2 Ex. 1(b–d) (call and put prices on the project, the parity identity; PS2 p. 1); lista-r3 #7 (Ross Cap. 16 Ex. 18, p. 578), #11 (Grinblatt–Titman Ex. 14.2–14.3, pp. 526–527), #15 (Amaro 2.1.5 #3–#4, p. 48).

2A.9–2A.10 · MM II, the WACC, and why r_D is neither the coupon nor the yield

> **Claim.** MM I fixes the value of the assets and their cash flows, hence r_A. Since r_A is the value-weighted average of r_E and r_D, more of the "cheap" debt must make equity dearer: $r_E = r_A + \frac{D}{E}(r_A - r_D)$ (MM II). It is linear in D/E while debt is safe and bends — in a two-state economy, goes flat — once r_D starts rising. The WACC is r_A and does not move. The r_D in these formulas is an **expected** return; a risky bond's promised yield overstates it.

Prev: L02-06-mm1-arbitrage-put-call · Next: L02-08-formulas · Map: L02-00-map

Notation.

Symbol	Meaning	Range
$r_A = r_{WACC} = r_U$	expected return on assets = WACC = return on unlevered equity	—
r_E, r_D	expected returns on levered equity and on debt	—
$x = D/E$	market debt-to-equity ratio	≥ 0
$V_L = D_L + E_L$, $V_U = E_U$	value of the levered and of the unlevered firm	> 0
F	face value of one-period debt	≥ 0
y	yield to maturity: $D = F/(1+y)$	—
p, C	default probability; expected recovery in default, $C < F$	$p \in [0,1]$
p_1, p_2	boom / slump state prices of the entrepreneur economy	0.357, 0.595

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1 · C — MM II, derived (slides 97–99)

Puzzle (slide 97): if the mix is irrelevant, why does debt require a lower return than equity? **Answer:** issuing debt makes equity riskier and so more costly — **even riskless debt** does (the entrepreneur example): it is not (only) about default.

MM I says A is independent of D, E; the cash flows $\tilde{C}_A$ are too (benchmark assumption 4); so from $A = E[\tilde{C}_A]/(1 + r_A)$ the return r_A is independent of D, E (slide 98). Start from the identity (1.10) (slide 99):

$$r_A = \frac{E}{E+D}\, r_E + \frac{D}{E+D}\, r_D \tag{7.1}$$

Multiply by $(E+D)/E$:

$$\frac{E+D}{E}\, r_A = r_E + \frac{D}{E}\, r_D$$

Write $\frac{E+D}{E} r_A = r_A + \frac{D}{E} r_A$ and solve for r_E:

$$\boxed{r_E = r_A + \frac{D}{E}\,(r_A - r_D)} \tag{7.2}$$

MM II (slide 97). *A firm's cost of equity increases as the market value debt-to-equity ratio increases; for low levels of debt, debt is risk-free and the increase is linear.* Since r_A is fixed and $r_A > r_D$, r_E rises with D/E; with $r_D = r_f$ fixed the slope is the constant $r_A - r_f$.

Same result in the books: Ross eq. [16.1] $R_E = R_A + (R_A - R_D) \times (D/E)$ (p. 548); Berk eq. 16.3 $r_E = r_U + \frac{D}{E}(r_U - r_D)$ (p. 471); Amaro de Matos **Proposition 2.1.2 (MM2)**, $R_s = R_s^u + (R_s^u - R_b)\frac{B}{S}$ (eq. 2.4, p. 43), derived from $R_s = E[V_j - B_j]/S = R_s^u \frac{V}{S} - R_b \frac{B}{S}$.

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2 · C — When debt turns risky (slide 100)

Once D is large enough to default with positive probability, r_D also rises with $x = D/E$. Differentiate (7.2), $r_E = r_A + x\,(r_A - r_D(x))$, with r_A constant (product rule on the second term):

$$\frac{\partial r_E}{\partial x} = (r_A - r_D) - x\,\frac{\partial r_D}{\partial x} \tag{7.3}$$

Read the slide's display with care. Slide 100 prints $\frac{\partial r_E}{\partial (D/E)} = r_A - \frac{\partial r_D}{\partial (D/E)}$; differentiating (7.2) gives (7.3), with $-r_D$ in the first term and the factor D/E on the second. The conclusion is the slide's: the two effects can offset, and *"with a finite number of states … r_E is strictly increasing in D/E and then it becomes flat while r_D increases."*

Worked: the flat region in the entrepreneur economy (built here on slides 55–56: $X_s = 1.05$, $X_b = 3.15$, probabilities $1/2$, $p_1 = 1/2.8$, $p_2 = 1/1.05 - 1/2.8$). Let the face be $F \in (1.05,\ 3.15)$: debt is paid in the boom and defaults in the slump, taking the slump's 1.05.

$$D(F) = p_1 F + p_2(1.05) = \frac{F}{2.8} + 0.625 \tag{7.4}$$

Equity pays $3.15 - F$ in the boom and 0 in the slump:

$$E(F) = p_1(3.15 - F) = \frac{3.15 - F}{2.8} \tag{7.5}$$

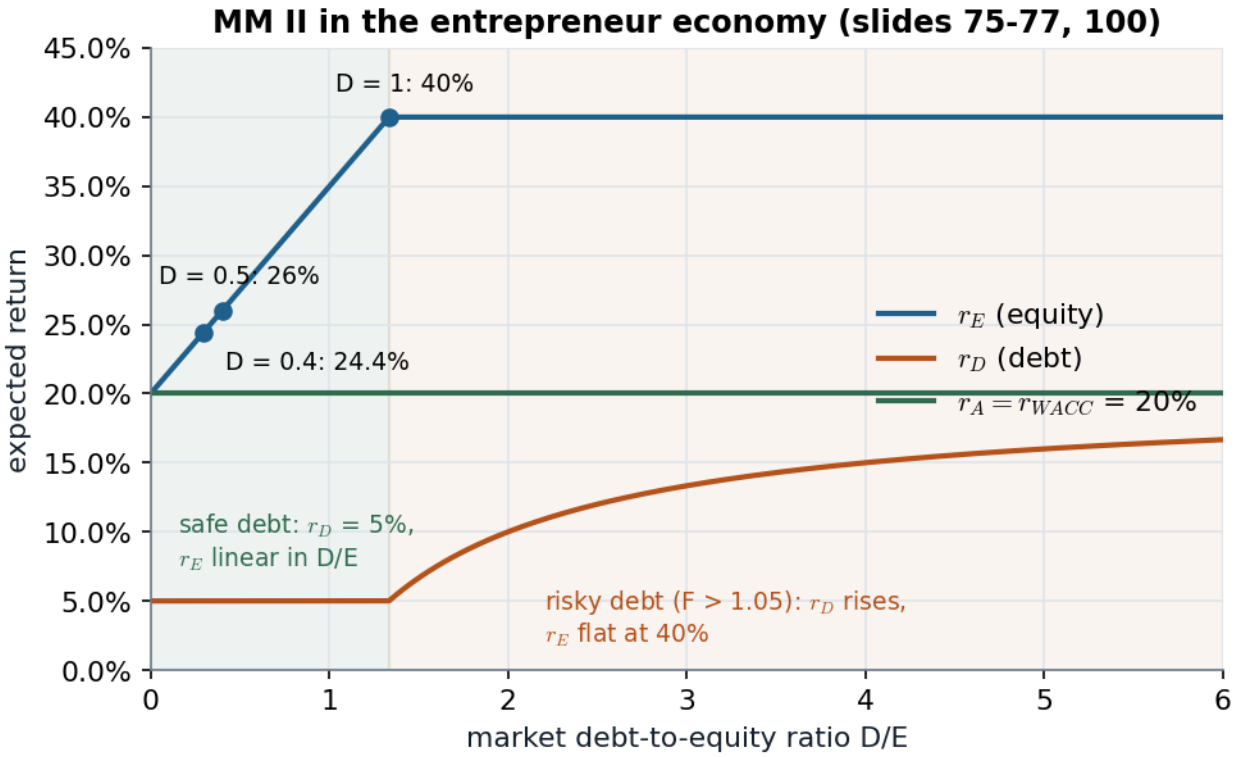
Its expected payoff is $\frac{1}{2}(3.15 - F)$, so

$$r_E = \frac{\frac{1}{2}(3.15 - F)}{(3.15 - F)/2.8} - 1 = \frac{2.8}{2} - 1 = 40\%, \qquad r_D = \frac{\frac{1}{2}F + \frac{1}{2}(1.05)}{F/2.8 + 0.625} - 1 \tag{7.6}$$

r_E is 40% for every risky face value: equity is now a pure boom claim, and a boom claim returns $q/p_1 - 1 = 0.5 \times 2.8 - 1 = 40\%$ however many units of it one holds. r_D rises from $1.05/1 - 1 = 5\%$ at $F = 1.05$ toward $2.1/1.75 - 1 = 20\% = r_A$ as $F \to 3.15$ (debt becomes the whole firm). At $F = 2.1$: $D = 0.75 + 0.625 = 1.375$, $E = 1.05/2.8 = 0.375$, $x = 3.667$, $r_D = 1.575/1.375 - 1 = 14.55\%$; check (7.2):

$$r_E = 20\% + 3.667\,(20\% - 14.55\%) = 20\% + 3.667 \times 5.45\% = 20\% + 20\% = 40\%\ \checkmark$$

and (7.3) is zero there: $x\,\partial r_D/\partial x = r_A - r_D$ exactly.



MM II (7.2) in both regions: linear while debt is safe (slides 75–77), flat once it is risky (7.6, slide 100); $r_A = r_{WACC} = 20\%$ *throughout.*

The kink at $D/E = 1.33$ is the face value 1.05 that exhausts the slump payoff. To its right the WACC still averages to 20%: equity's weight shrinks while debt's return climbs toward 20%.

> **Interactive companion.** The State-Price Balance Sheet (`companions/companion-mm-state-prices.html`) — drag F past the slump payoff: the right panel's r_E stops rising exactly where r_D leaves r_f; press "Risky debt, F = 2.1" for the case above.

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3 · C — The WACC and conservation of value (slides 101–102)

r_A **is** the weighted average cost of capital, r_{WACC}, and by MM I it does not depend on D, E. Another statement of MM I:

$$D_L + E_L = V_L = V_U = E_U \tag{7.7}$$

value of the levered firm (any D_L, E_L) = value of the unlevered, all-equity firm. Hence

$$r_A = r_{WACC} = \frac{E}{D+E}\,r_E + \frac{D}{D+E}\,r_D = r_U \tag{7.8}$$

and (7.2) becomes

$$r_E = r_U + \frac{D}{E}\,(r_U - r_D) \tag{7.9}$$

Slide 102's first display swaps the weights — it prints $\frac{D}{D+E}r_E + \frac{E}{D+E}r_D$. The correct weights are (7.8)'s: equity's return times equity's share.

Conservation of value (slide 102): *"With perfect capital markets, financial transactions neither add nor destroy value, but instead represent a repackaging of risk (and therefore return)."*

U-layer, Ross Exemplo 16.3 (p. 549): WACC (no taxes) 12%, borrowing at 8%. Target 20% debt, $D/E = 0.2/0.8 = 0.25$:

$$R_E = 12\%+(12\%-8\%)\times 0.25 = 12\%+1\% = 13\%; \qquad \text{CMPC} = 0.8(13\%)+0.2(8\%) = 10.4\%+1.6\% = 12\%$$

At 50% debt, $D/E = 1$: $R_E = 12\% + 4\% = 16\%$ and $0.5(16\%) + 0.5(8\%) = 12\%$.

U-layer, Berk Fig. 16.5 (p. 471): the coffee shop's WACC stays at 15% for $D = 0,\ 6{,}000,\ 15{,}000,\ 27{,}000$ while r_E goes $15\% \to 17.5\% \to 25\% \to 75\%$ and r_D rises to 8.3% at the last level, where debt has become risky: $75\% \times 0.1 + 8.33\% \times 0.9 = 7.5\% + 7.5\% = 15\%$.

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4 · C — r_D is not the coupon, and not the yield (slides 103–105)

r_D is the **expected return investors require today** on the firm's debt. It is not the interest rate contracted in the past: macroeconomic conditions and the firm's fundamentals (default risk) change (slide 103). It is not the yield either. For one-period debt of face F:

$$D = \frac{F}{1+y} \qquad \text{(yield: discounts the promised payment)} \tag{7.10}$$

$$D = \frac{F(1-p)+C\,p}{1+r_D} \qquad \text{(expected return: discounts the expected payment)} \tag{7.11}$$

Equate (7.10) and (7.11) and solve:

$$\frac{1+y}{1+r_D} = \frac{F}{F(1-p)+Cp}$$

With $p > 0$ and $C < F$, $F(1-p)+Cp = F - p(F-C) < F$, so the right side exceeds 1:

$$\boxed{y > r_D \quad \text{whenever debt is risky}} \tag{7.12}$$

The yield overestimates the debt cost of capital (slide 105). Using it in (7.8) overstates the WACC.

Worked (the $F = 2.1$ case of §2). $D = 1.375$; default in the slump ($p = 0.5$) with recovery $C = 1.05$:

$$y = \frac{2.1}{1.375} - 1 = 52.73\%$$

$$r_D = \frac{2.1(1-0.5)+1.05(0.5)}{1.375} - 1 = \frac{1.05+0.525}{1.375} - 1 = \frac{1.575}{1.375} - 1 = 14.55\%$$

The 38-point gap is expected default loss. The remaining $14.55\% - 5\%$ over r_f is a **risk premium**: the debt defaults in the slump, the expensive state, so it carries systematic risk.

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5 · G — The beta form of MM II (Amaro de Matos p. 43)

Linearity of covariance turns (7.2) into $\beta_s = \beta_s^u(1 + \frac{B}{S}) - \beta_d \frac{B}{S}$, and the average cost of capital $\bar{R} = \frac{S}{S+B}R_s + \frac{B}{S+B}R_b$ has beta $\bar{\beta} = \frac{S}{S+B}\beta_s + \frac{B}{S+B}\beta_d$ — the relevering formula (1.12) of L02-01-present-value-capm-betas with risky debt.

Links

- The identity behind (7.1) → L02-01-present-value-capm-betas §5; the safe-debt tables → L02-05-entrepreneur-example §6–7; MM I → L02-06-mm1-arbitrage-put-call.
- With taxes the WACC falls in D/V and $V_L = V_U + \tau D$ → L03; ROE and P/E moving with leverage → L01.
- Rules: rules/03 — MM benchmark.
- Python: `mm2_rE`, `yield_vs_rD` → L02-10-python.
- **Practice.** PS2 Ex. 1(e) (r_D, r_E, r_A across D; PS2 p. 1); PS2 Ex. 4(a–b) (beta and expected return after a debt-financed repurchase; pp. 2–3); lista-r3 #5 (Berk Ch. 16 Ex. 16.5, p. 491); lista-r2 #4 (Berk Ch. 13 Ex. 4, p. 403).

L02 · Formulas — the exam specification, every equation, every number

Prev: L02-07-mm2-wacc-cost-of-debt · Next: L02-09-pitfalls · Map: L02-00-map

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0 · The specification, mapped

The slides state no learning goals; the specification is the lecture's block list (Map/lectures-index, Topic 2A), item by item.

Block	Slides	What must be done with it	Note
2A.1 Present value	2–7	write P_A as an expected PV; distinguish expected and realized return; price a coupon bond with spot rates	L02-01-present-value-capm-betas §1
2A.2 Discount rates and betas	8–23	apply the CAPM; explain why a levered comparable's β_E overstates β_A; **derive** r_A and β_A as value-weighted averages; unlever comparables and compute the project rate	L02-01-present-value-capm-betas §2–6
2A.3 Which cash flows	24–27	build unlevered FCF; incremental basis; opportunity in, sunk out; consistent inflation; terminal and after-tax salvage value; NWC recovery	L02-02-cash-flows-npv-irr §1–2
2A.4 NPV rule	28–38	state the rule and its "arbitrage across the boundary" reading; show NPV = shareholders' gain under three financings; list and exhibit the IRR pitfalls	L02-02-cash-flows-npv-irr §3–5
2A.5 Debt vs. equity	39–45	the four differences; covenants, technical default, limited liability; payoffs min / max; equity as a call	L02-03-debt-vs-equity-claims

Block	Slides	What must be done with it	Note
2A.6 Benchmark, complete markets	46–54	the assumptions; contingent claims, completeness, law of one price, the two arbitrages; compute state prices by replication	L02-04-complete-markets-state-prices
2A.7 Entrepreneur example	55–81	state prices; A, D, E, α under each financing; returns tables; r_E as a function of D; spot the "1.77" fallacy	L02-05-entrepreneur-example
2A.8 MM I	82–96	prove MM I by arbitrage both ways (homemade leverage); risky debt = bond − put, equity = call; parity ⇒ $V(K)$ constant	L02-06-mm1-arbitrage-put-call
2A.9 MM II and WACC	97–102	derive MM II; linear vs. flat region; $r_A = r_{WACC} = r_U$; conservation of value	L02-07-mm2-wacc-cost-of-debt §1–3
2A.10 r_D vs. coupon and yield	103–105	define y and r_D; prove $y > r_D$ for risky debt	L02-07-mm2-wacc-cost-of-debt §4

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1 · Every numbered equation

Eq.	Statement	Derived in
(1.1)	$P_A = \sum_t E[\tilde{d}_t]/(1+r_t)^t$	L02-01-present-value-capm-betas §1
(1.2)	coupon bond with spot rates	§1
(1.3)	constant rate: $P_A = \sum_t E[\tilde{d}_t]/(1+r)^t$	§1
(1.4)	CAPM $r = r_f + \beta(E[\tilde{r}_m] - r_f)$	§2
(1.5)	$\beta = \mathrm{Cov}(\tilde{r}, \tilde{r}_m)/\mathrm{Var}(\tilde{r}_m)$	§2
(1.6)	project rate from a comparable, $\beta = \beta_A$	§3
(1.7)–(1.8)	house: $E[\tilde{r}_m] = 7\%$, Var = 0.0036	§4
(1.9)	$D = E[\tilde{C}_D]/(1+r_D)$, etc.	§5
(1.10)	$r_A = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D$ (also realized returns)	§5

Eq.	Statement	Derived in
(1.11)	$\beta_A = \frac{E}{D+E}\beta_E + \frac{D}{D+E}\beta_D$	§5
(1.12)	$r_E = r_A + \frac{D}{E}(r_A - r_D)$; $\beta_E = \beta_A + \frac{D}{E}(\beta_A - \beta_D)$	§5
(2.1)	$\text{FCF} = \text{EBIT}(1-\tau) + \text{Dep} - \text{CAPEX} - \Delta\text{NWC}$	L02-02-cash-flows-npv-irr §1
(2.2)	after-tax salvage $SV - \tau(SV - \text{PPE})$	§2
(2.3)	$\text{TV}_T = \text{FCF}_{T+1}/(r-g)$, $g < r$	§2
(2.4)	$NPV_A = PV_A - c_A$	§3
(2.5)	slides' project: $-15 + 22/1.1 = 5$	§4
(2.6)	IRR: $0 = -c_A + \sum_t E[\tilde{d}_t]/(1+\text{IRR})^t$	§5
(3.1)–(3.2)	$\tilde{C}_D = \min(\tilde{A}, F)$, $\tilde{C}_E = \max(\tilde{A} - F, 0)$	L02-03-debt-vs-equity-claims §2
(3.3)	$\tilde{C}_D + \tilde{C}_E = \tilde{A}$	§2
(3.4)	$\min(\tilde{A}, F) = F - \max(F - \tilde{A}, 0)$	§2
(4.1)–(4.4)	replication of cc_1, cc_2; $p_1 = 0.381$, $p_2 = 0.571$	L02-04-complete-markets-state-prices §3
(4.5)	price $= 0.381y + 0.571x$ (linear pricing)	§3
(4.6)	$\pi_s = p_s / \sum p$; price $= \frac{1}{1+r_f}\sum \pi_s \text{ payoff}_s$	§4
(5.1)–(5.4)	entrepreneur: $p_1 = 1/2.8 = 0.357$, $p_2 = 1/1.05 - 1/2.8 = 0.595$	L02-05-entrepreneur-example §2
(5.5)–(5.6)	entrepreneur under debt $2.1p_1 = 0.75$; $A = 3.15p_1 + 1.05p_2 = 1.75$	§3
(5.7)–(5.9)	$1 = \alpha(3.15p_1 + 1.05p_2)$, $\alpha = 4/7$; entrepreneur $\frac{3}{7}(1.75) = 0.75$	§4
(5.10)–(5.12)	$r_E = \frac{E+D}{E}r_A - \frac{D}{E}r_D = 20\% + \frac{D}{E}15\% = 20\% + \frac{D}{1.75-D}15\%$	§7
(6.1)–(6.2)	Step 1 ($V_2 > V_1$): cost 0, payoff $\alpha X(V_2/V_1 - 1) > 0$	L02-06-mm1-arbitrage-put-call §2
(6.3)–(6.4)	Step 2 ($V_1 > V_2$): cost 0, payoff $\alpha X(V_1/V_2 - 1) > 0$	§2
(6.5)–(6.6)	$\min\{K, X\} = K - \max\{K - X, 0\}$; $D = K/(1+r_f) - P(K)$	§3
(6.7)–(6.8)	$E = C(K)$; $V(K) = C(K) + K/(1+r_f) - P(K)$, independent of K	§4
(7.1)–(7.2)	MM II: $r_E = r_A + \frac{D}{E}(r_A - r_D)$	L02-07-mm2-wacc-cost-of-debt §1
(7.3)	$\partial r_E/\partial(D/E) = (r_A - r_D) - \frac{D}{E}\,\partial r_D/\partial(D/E)$	§2

Eq.	Statement	Derived in
(7.4)–(7.6)	risky face $F \in (1.05, 3.15)$: $D = F/2.8 + 0.625$, $E = (3.15 - F)/2.8$, $r_E = 40\%$	§2
(7.7)–(7.9)	$V_L = V_U$; $r_A = r_{WACC} = r_U$; $r_E = r_U + \frac{D}{E}(r_U - r_D)$	§3
(7.10)–(7.12)	$D = F/(1+y)$; $D = [F(1-p) + Cp]/(1+r_D)$; $y > r_D$ if risky	§4

1.1 · Named results

Result	Statement	Note
NPV rule	undertake iff $NPV > 0$; frictionless + all-equity ⇒ shareholder gain = NPV	L02-02-cash-flows-npv-irr §3–4
Law of one price / no arbitrage	same payoff ⇒ same price; no free lunch, no money machine	L02-04-complete-markets-state-prices §2
MM I (slide 83)	in a frictionless economy, total firm value is independent of capital structure	L02-06-mm1-arbitrage-put-call
MM II (slide 97)	cost of equity rises with market D/E; linear while debt is safe	L02-07-mm2-wacc-cost-of-debt
Conservation of value (slide 102)	financial transactions repackage risk; they neither add nor destroy value	L02-07-mm2-wacc-cost-of-debt §3
Fisher separation (Amaro Prop. 1.1.1, p. 12) — G	investment decided by the market rate, independent of preferences	L02-02-cash-flows-npv-irr §6
Amaro Prop. 1.2.1 (p. 22) — G	no arbitrage ⇔ strictly positive state prices	L02-04-complete-markets-state-prices §5
Amaro Prop. 2.1.1 MM1 (p. 42), Prop. 2.1.2 MM2 (p. 43) — G	$V_l = V_u$; $R_s = R_s^u + (R_s^u - R_b)B/S$	notes 06, 07

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2 · Every number in the lecture

Case	Inputs	Results	Note
House (slides 12–15)	100; market +10% (0.8) / −5% (0.2); r_f 2%	$E[\tilde{r}_m]$ 7%, Var 0.36%; 20% debt: r_E **8.25%** (slide: 8%), β_E 1.25; 50% debt: 12%, 2	01 §4
Comparables (22–23)	r_f 4%, $E[\tilde{r}_m]$ 12.4%; β_E 0.75 / 1 / 1.08	β_A 0.72 / 0.77 / 0.85; mean 0.78 (0.7811); r **10.56%**	01 §6

Case	Inputs	Results	Note
Polaris (Higgins pp. 303, 319)	β_E 2.05; E/V 5,898/7,334	K_E 14.8%; β_A 1.65	01 §2, §6
NPV and shareholders (32–36)	assets 100/10; project $-15 \to 22$; r_f 10%	NPV 5; equity $50 \to 70$; $\alpha = 15/70$, old holders 55; debt face 16.5; gain 5 three ways	02 §4
IRR (38)	A $-1000/+1500$, B $+1000/-1500$; A $-10/14.3$, B $-1000/1210$	IRR 50%/50%, NPV ± 363.6; IRR 43%/21%, NPV 3/100	02 §5
Mine (Ross pp. 297–298)	-60, $+155$, -100	IRR 25% and 33⅓%; NPV(10%) -1.74	02 §5
Stock/bond (52–54)	stock $2 \to (1.9, 2.4)$; bond $1 \to 1.05$	p_{boom} 0.381, p_{slump} 0.571; $\pi = 0.4$ / 0.6	04 §3–4
Entrepreneur (55–80)	project $1 \to (1.05, 3.15)$; stock $1 \to (0, 2.8)$; bond $1 \to 1.05$	p_1 0.357, p_2 0.595; A **1.75**; NPV **0.75**; α 4/7; r_A **20%**	05
— debt 1	F 1.05	D 1, E 0.75; r_D 5%, r_E 40%	05 §3, §6
— debt 0.4	F 0.42	E 1.35, r_E 24.4%	05 §7
— debt 0.5	F 0.525	E 1.25, r_E 26% (fallacy: 1.27, A 1.77)	05 §7
— risky debt (built here)	F 2.1	D 1.375, E 0.375, r_D 14.55%, y 52.73%, r_E 40%	07 §2, §4
Coffee shop (Berk pp. 465–471)	24,000 → 34,500; 15%	V^U 30,000; NPV 6,000; r_E 25% (D 15,000), 17.5% (D 6,000), 75% (D 27,000)	05 §8, 07 §3
Ricardo S/A (Ross p. 549)	R_A 12%, R_D 8%	R_E 13% at D/E 0.25; 16% at 1	07 §3

To know cold: $A = 1.75$, NPV $= 0.75$, $r_A = 20\%$, $p_1 = 0.357$, $p_2 = 0.595$, $r = 10.56\%$.

L02 · Pitfalls and a True/False self-test

Prev: L02-08-formulas · Next: L02-10-python · Map: L02-00-map

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Part I — the confusions this lecture produces

1 · Book value vs. market value of the assets

The entrepreneur invests 1 and expects 2.1: "$r_A = 110\%$". **Wrong denominator.** The market value is $A = 1.75$, so $r_A = 2.1/1.75 - 1 = 20\%$ (slide 55). The 0.75 gap is the NPV. Returns, betas and leverage ratios in this lecture are always on **market** values. → L02-05-entrepreneur-example §1.

2 · Equity beta vs. asset beta

Plugging a levered comparable's β_E into the CAPM charges the project for leverage it does not have. Minimal counterexample: the house has $\beta_A = 1$; financed 50/50 its equity has $\beta_E = 2$ (slide 15). Averaging the comparables' **equity** betas gives 11.92% instead of 10.56%. → L02-01-present-value-capm-betas §4–6.

3 · "Debt is cheaper, so borrowing lowers the cost of capital"

Holding r_E at 24.4% while raising D from 0.4 to 0.5 makes A jump to 1.77 (slide 79). The correct r_E is 26% and A stays 1.75. **Every unit of cheap debt makes the remaining equity dearer by exactly enough** (MM II). → L02-05-entrepreneur-example §7.

4 · "Equity gets riskier only because of default"

In the entrepreneur example debt is **riskless** for every $D \leq 1$, yet r_E rises from 20% to 40% (slides 75–77, 97). Leverage concentrates the assets' **systematic** risk on a smaller equity base; default risk is a second, separate effect. → L02-07-mm2-wacc-cost-of-debt §1.

5 · "MM II says r_E is linear in D/E"

Only while r_D is constant. With risky debt (7.3) holds and, with finitely many states, r_E can go **flat** (40% for every face above 1.05 in the entrepreneur economy) while r_D rises. And it is linear in D/E, **not** in D: $r_E = 20\% + 15\%\, D/(1.75 - D)$ is convex in D (slide 76). → L02-07-mm2-wacc-cost-of-debt §2.

6 · The yield as the cost of debt

With $F = 2.1$, the yield is 52.73% and r_D is 14.55%: the yield discounts the **promised** payment, r_D the **expected** one (slides 104–105). Putting y in a WACC overstates it. Nor is r_D the coupon contracted years ago (slide 103). → L02-07-mm2-wacc-cost-of-debt §4.

7 · State prices vs. probabilities

Both states have probability ½, yet $p_1 = 0.357 \neq p_2 = 0.595$ (slides 65–67). State prices embed time **and** risk aversion; the risk-neutral weights π are not beliefs either (0.4 / 0.6 in the stock–bond example). → L02-04-complete-markets-state-prices §4.

8 · IRR as a ranking device

Same IRR (50%), opposite NPVs (±363.6) for lending vs. borrowing; higher IRR (43% vs. 21%) for the project that creates 3 instead of 100 (slide 38); two IRRs above the cost of capital with a negative NPV (Ross' mine). → L02-02-cash-flows-npv-irr §5.

9 · Putting the tax shield in the cash flow

(2.1) taxes EBIT as if unlevered on purpose: financing subsidies are counted **in the discount rate** (slide 25). Adding interest tax savings to FCF *and* using an after-tax WACC counts them twice. → L02-02-cash-flows-npv-irr §1.

10 · Two slide displays to read with care

Slide 100 prints $\partial r_E/\partial(D/E) = r_A - \partial r_D/\partial(D/E)$; differentiating MM II gives $(r_A - r_D) - \frac{D}{E}\partial r_D/\partial(D/E)$. Slide 102's first display swaps the WACC weights ($\frac{D}{D+E}$ on r_E). The slide 14 table rounds r_E to 8% (exact 8.25%), and slide 23 rounds $\bar{\beta}_A$ to 0.78 while computing 10.56% from 0.7811.

Part II — True/False self-test (original items)

Scoring as in the exams: $\max\{0,\ \text{right} - \text{wrong}/2\}$; blanks do not count. **[E]** easy · **[M]** medium · **[H]** hard.

#	Claim	Verdict
1	If a project's cash flows have $\beta = 0$, its discount rate is r_f even if they are very volatile. **[E]**	**T**
2	A firm whose equity beta is 1.5 and whose debt is riskless has an asset beta above 1.5. **[E]**	**F**
3	$r_A = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D$ requires the CAPM to hold. **[M]**	**F**
4	Averaging comparables' equity betas and then unlevering at the average leverage gives the same answer as averaging their asset betas. **[H]**	**F**
5	Unlevered free cash flow includes the interest tax shield, since it is cash the firm keeps. **[E]**	**F**
6	A consultancy fee paid last month for a project study belongs in the project's NPV. **[E]**	**F**
7	A positive-NPV project is an arbitrage that competition should eliminate, so in equilibrium no firm has one. **[M]**	**F**

#	Claim	Verdict
8	In a frictionless all-equity firm, the shareholders' gain from a positive-NPV project is the same whether it is funded by insiders, new shares or safe debt. [**E**]	**T**
9	Two projects with the same IRR have the same NPV at any given discount rate. [**E**]	**F**
10	A project whose NPV profile never crosses zero has no IRR, and the NPV rule still decides it. [**M**]	**T**
11	Equity's payoff $\max(\tilde{A} - F, 0)$ is convex in $\tilde{A}$, so shareholders can gain from a mean-preserving increase in asset risk. [**M**]	**T**
12	Risky debt is worth a riskless bond of the same face value plus a put on the firm's assets. [**M**]	**F**
13	With two states and two securities with linearly independent payoffs, the market is complete. [**E**]	**T**
14	Equally likely states must have equal state prices in an arbitrage-free market. [**E**]	**F**
15	The sum of the state prices equals $1/(1 + r_f)$. [**E**]	**T**
16	In the entrepreneur example, the entrepreneur keeps more NPV under equity financing because she does not have to repay anything. [**E**]	**F**
17	In the entrepreneur example, r_A is 110% because the project turns an investment of 1 into an expected 2.1. [**E**]	**F**
18	MM I's arbitrage proof needs individuals to borrow at the same rate as firms. [**M**]	**T**
19	If a levered firm traded **above** its unlevered twin, an investor could profit by buying the levered firm's debt and equity in proportion and shorting the twin. [**M**]	**F**

#	Claim	Verdict
20	Put–call parity implies that the value of a firm with risky debt does not depend on the face value of that debt. **[M]**	**T**
21	Under MM, borrowing to repurchase shares raises the expected return on equity and leaves the WACC unchanged. **[E]**	**T**
22	MM II implies r_E rises without bound as D/E grows, even when debt becomes risky. **[H]**	**F**
23	Riskless debt cannot raise the equity beta, because it carries no risk. **[E]**	**F**
24	For risky one-period debt, the promised yield exceeds the expected return on the debt. **[M]**	**T**
25	r_D can be read off the coupon rate written in the firm's existing loan contracts. **[E]**	**F**

Justifications

1. **T.** The CAPM prices only covariance with the market: $\beta = 0$ in (1.4) gives $r = r_f$. Idiosyncratic volatility is diversifiable and unpriced (slide 31's alternative assumption).
2. **F.** With $\beta_D = 0$, $\beta_A = \frac{E}{D+E}\beta_E \leq \beta_E$ by (1.11): unlevering lowers the beta. It equals 1.5 only if $D = 0$.
3. **F.** (1.10) follows from $A = D + E$ in market value and $\tilde{C}_A = \tilde{C}_D + \tilde{C}_E$ alone (slides 17–19); the CAPM only enters when one maps r's into β's.
4. **F.** Unlevering is nonlinear in leverage across firms. Slide-23 data: the mean β_E is 0.943 and the mean $E/(D+E)$ is $(0.96 + 0.77 + 0.79)/3 = 0.84$, giving $0.943 \times 0.84 = 0.792$, not the 0.781 average of asset betas. Close here, but not the same object.
5. **F.** (2.1) taxes EBIT as if the firm were unlevered; financing subsidies go into the discount rate (slide 25). Including them in the cash flow double counts with an after-tax WACC.
6. **F.** It is sunk: paid whether or not the project goes ahead (slide 26; Ross p. 320). Only cash flows that differ *with* vs. *without* the project count.
7. **F.** Slide 29: unlike financial arbitrages, positive-NPV projects are **not scalable**, so they can exist in equilibrium; they are rents to a specific capability or position.
8. **T.** Slides 33–36: gain 5 in each case. Competitive investors take zero NPV, so the whole NPV stays with the old shareholders. Fails with risky debt or asymmetric information (slide 31).
9. **F.** Slide 38: A and B both have IRR 50% and NPVs +363.6 and −363.6 at 10%. The IRR ignores whether the flows are lending or borrowing, and their scale.
10. **T.** Example $+100, -300, +250$ (L02-02-cash-flows-npv-irr §5): the discriminant is negative, NPV > 0 at every rate; the NPV rule says accept.
11. **T.** A convex payoff's expectation rises with a mean-preserving spread (Jensen): with $F = 1.05$,

spreading (1.05, 3.15) to (0, 4.2) raises equity's mean from 1.05 to 1.575 and lowers debt's from 1.05 to 0.525 (L02-03-debt-vs-equity-claims §3; Tirole p. 77).

12. **F.** Minus a put: $\min(K, X) = K - \max(K - X, 0)$ (6.5), so $D = K/(1 + r_f) - P(K)$ (slide 92). The put is the creditors' loss from limited liability.
13. **T.** Completeness ⇔ number of linearly independent payoff vectors = number of states (Amaro p. 21); the stock (0, 2.8) and bond (1.05, 1.05) are an example (slides 64–66).
14. **F.** $p_1 = 0.357$ and $p_2 = 0.595$ with probabilities ½ each (slide 67): risk aversion makes a unit in the bad state dearer. No-arbitrage only requires $p_s > 0$.
15. **T.** A riskless unit is one of each contingent claim: $\sum_s p_s = 1/(1 + r_f)$; $0.357 + 0.595 = 0.952 = 1/1.05$.
16. **F.** She keeps 0.75 under debt (5.5), under equity (5.9) and under any mix (slide 70). Selling 4/7 of the firm for 1 is a fair trade, not a free one.
17. **F.** 1 is the cost (book value). r_A is on the market value: $2.1/1.75 - 1 = 20\%$ (slides 55, 71).
18. **T.** Step 2 (slide 89) levers up the unlevered twin with a **personal** loan at R_f (assumption 2). With a higher personal rate, the arbitrage closes only within a band.
19. **F.** Wrong direction. Buying the levered firm's debt and equity in proportion and shorting the twin is Step 1 (slide 87), profitable when the **twin** is dearer, $V_2 > V_1$. If the levered firm is dearer, $V_1 > V_2$, the trade is Step 2 (slide 89): short its equity, buy the twin and borrow personally — homemade leverage.
20. **T.** $V(K) = C(K) + K/(1 + r_f) - P(K)$ replicates X for every K (slides 94–96), so it equals the asset's price whatever K.
21. **T.** MM II raises r_E by $\frac{D}{E}(r_A - r_D)$; MM I fixes $r_A = r_{WACC}$ (slides 99–101).
22. **F.** Once debt is risky, r_D rises and (7.3) can turn zero: in the two-state economy r_E is flat at 40% for every face above 1.05 (slide 100).
23. **F.** Slide 97: even risk-free debt makes equity riskier. The house: riskless debt of 50 takes β_E from 1 to 2 (slide 15).
24. **T.** (7.12): with $p > 0$ and $C < F$ the expected payment is below the promised one, so the rate that discounts the promise to D is higher (slide 105).
25. **F.** r_D is today's required expected return; macro conditions and default risk change after the contract is signed (slide 103).

L02 · Python — every routine the lecture needs, checked against its own numbers

Prev: L02-09-pitfalls · Map: L02-00-map

What exists already, and why it is cited rather than imported. `Listas/lista-r2-valuation-solutions_check.py` and `Listas/lista-r3-mm-taxes-tradeoff-solutions_check.py` check every number of the list solutions (run them with `python -I <path>`; each prints its PASS count). They are flat scripts — every check runs at import, and the hyphenated file names are not importable — so their `npv` / `root` helpers are re-stated here in a few lines instead. Their coverage complements this note: lista-r2 #6 (unlevering comparables), #8 (multiple IRRs), #14 (state prices from a traded project); lista-r3 #1 (two-outcome MM), #7 (homemade leverage), #11 (risky debt in a binomial market). The figures' own checks are in `Notas/L02/figures/make_figures.py`; the companion's in `Notas/L02/companions/check_companions.py`.

Run. Copy the block into a file and run it with `python -I`; it prints `L02 python: all checks pass` or stops at the first failing `assert`. numpy only.

```
"""L02 reference implementation. Each routine: purpose, inputs, failure mode in its
↪ docstring;
asserts use the slides' (or the cited book's) printed numbers."""
import numpy as np


# ---------- 2A.1-2A.4: present value, NPV, IRR ----------
def npv(r, cfs):
    """NPV at rate r of cash flows cfs[0..T], cfs[0] at t = 0 (eqs. 1.3, 2.4).
    r > -1 required (else division by a non-positive discount factor)."""
    return sum(c / (1 + r) ** t for t, c in enumerate(cfs))


def irr_roots(cfs, lo=-0.99, hi=5.0, n=20000):
    """All IRRs in (lo, hi): sign changes of NPV on a grid, refined by bisection (eq.
    ↪ 2.6).
    Returns [] when the NPV never crosses zero (no IRR) - the caller must handle
↪ that.
    A double root (NPV tangent to zero) is missed: grid sign-change method."""
    grid = np.linspace(lo, hi, n)
    vals = [npv(r, cfs) for r in grid]
    roots = []
    for a, b, fa, fb in zip(grid, grid[1:], vals, vals[1:]):
        if fa == 0:
            roots.append(a)
        elif fa * fb < 0:
            for _ in range(100):
                m = (a + b) / 2
                if npv(a, cfs) * npv(m, cfs) <= 0:
                    b = m
                else:
                    a = m
            roots.append((a + b) / 2)
```

```
    return roots


assert abs(-15 + 22 / 1.10 - 5) < 1e-12                                    # slide
↪  32
assert np.allclose(irr_roots([-60, 155, -100]), [0.25, 1 / 3], atol=1e-9)  # Ross pp.
↪  297-298
assert round(npv(0.10, [-60, 155, -100]), 2) == -1.74                      # Ross p.
↪  297
assert irr_roots([100, -300, 250]) == []                                   # no IRR
↪  (L02-02 s5)
assert np.allclose(irr_roots([-1000, 1500]), [0.5]) and np.allclose(irr_roots([1000,
↪  -1500]), [0.5])
assert round(npv(0.10, [-1000, 1500]), 1) == 363.6                         # slide
↪  38
assert round(npv(0.10, [-10, 14.3]), 6) == 3.0 and round(npv(0.10, [-1000, 1210]), 6)
↪  == 100.0


def tv_growing(fcf_next, r, g):
    """Growing-perpetuity terminal value FCF_{T+1}/(r-g) (eq. 2.3). Fails
    ↪  (ValueError) if g >= r."""
    if g >= r:
        raise ValueError("g must be below r for the perpetuity to converge")
    return fcf_next / (r - g)


assert abs(tv_growing(1.0, 0.10, 0.02) - npv(0.10, [0] + [1.02 ** k for k in
↪  range(3000)])) < 1e-9


# ---------- 2A.2: CAPM, leverage and betas ----------
def capm(rf, beta, rm):
    """Required expected return r_f + beta (E[r_m] - r_f) (eq. 1.4)."""
    return rf + beta * (rm - rf)


def unlever(beta_E, D, E, beta_D=0.0):
    """Asset beta as the value-weighted average (eq. 1.11); D, E market values, D + E
    ↪  > 0."""
    return E / (D + E) * beta_E + D / (D + E) * beta_D


def relever(beta_A, D, E, beta_D=0.0):
    """Equity beta beta_A + (D/E)(beta_A - beta_D) (eq. 1.12); E > 0."""
    return beta_A + D / E * (beta_A - beta_D)


def moments(x, y, prob):
    """Mean of x and Cov(x, y) under state probabilities prob (eq. 1.5)."""
```

```
    x, y, prob = map(np.asarray, (x, y, prob))
    return prob @ x, prob @ ((x - prob @ x) * (y - prob @ y))


# the house, slides 12-15
q, rm_s = np.array([0.8, 0.2]), np.array([0.10, -0.05])
Erm, var_m = moments(rm_s, rm_s, q)
assert np.isclose(Erm, 0.07) and np.isclose(var_m, 0.0036)
for debt, rE_print, bE_print in ((20, 0.0825, 1.25), (50, 0.12, 2.0)):
    eq_val = 100 - debt
    rE_s = (100 * (1 + rm_s) - debt * 1.02) / eq_val - 1
    ErE, cov = moments(rE_s, rm_s, q)
    assert np.isclose(ErE, rE_print) and np.isclose(cov / var_m, bE_print)   # slide
     ↪ 14 prints 8% (rounded)
    assert np.isclose(relever(1.0, debt, eq_val), bE_print)

# three comparables, slides 22-23
bA = [unlever(b, d, e) for b, d, e in ((0.75, 0.004, 0.096), (1.0, 2.3, 7.7), (1.08,
 ↪ 0.21, 0.79))]
assert np.allclose(np.round(bA, 2), [0.72, 0.77, 0.85])
assert round(np.mean(bA), 2) == 0.78 and round(100 * capm(0.04, np.mean(bA), 0.124),
 ↪ 2) == 10.56
assert round(unlever(2.05, 7334 - 5898, 5898), 2) == 1.65                  # Higgins
 ↪ p. 319
assert round(100 * capm(0.017, 2.05, 0.017 + 0.064), 1) == 14.8            # Higgins
 ↪ p. 303


# ---------- 2A.6-2A.7: state prices and the entrepreneur ----------
def state_prices(payoffs, prices):
    """State prices p solving payoffs @ p = prices (one row per security, one column
     ↪ per state).
    payoffs must be square and invertible (complete market, N = K); raises
 ↪ LinAlgError otherwise.
    A non-positive entry in the result signals an arbitrage (Amaro Prop. 1.2.1)."""
    p = np.linalg.solve(np.asarray(payoffs, float), np.asarray(prices, float))
    assert (p > 0).all(), "non-positive state price: the securities admit an
     ↪ arbitrage"
    return p


def price(payoff, p):
    """Law of one price: value = sum_s p_s payoff_s (eq. 4.5)."""
    return float(np.dot(payoff, p))


p_ex = state_prices([[1.9, 2.4], [1.05, 1.05]], [2, 1])                    # columns:
 ↪ slump, boom
assert np.allclose(np.round(p_ex, 3), [0.571, 0.381])                     # slides
 ↪ 53-54
```

```
assert np.allclose(p_ex / p_ex.sum(), [0.6, 0.4])                                #
 ↪ risk-neutral pi (4.6)

P = state_prices([[0, 2.8], [1.05, 1.05]], [1, 1])                              # [slump
 ↪ p2, boom p1]
p2, p1 = P
assert round(p1, 3) == 0.357 and round(p2, 3) == 0.595                          # slides
 ↪ 65-66
X = np.array([1.05, 3.15])                                                       # slump,
 ↪ boom (slide 55)
QE = np.array([0.5, 0.5])


def claims(F, X=X, P=P, prob=QE):
    """Values and expected returns of debt with face F and equity on payoff X (eqs.
     ↪ 3.1-3.3,
    5.6, 7.4-7.6). Returns dict A, D, E, rA, rD, rE; a return is None when its value
 ↪ is 0."""
    dpay, epay = np.minimum(X, F), np.maximum(X - F, 0)
    A, D, E = price(X, P), price(dpay, P), price(epay, P)
    ret = lambda pay, v: float(prob @ pay) / v - 1 if v > 1e-12 else None
    return dict(A=A, D=D, E=E, rA=ret(X, A), rD=ret(dpay, D), rE=ret(epay, E))


def alpha_equity(cash, X=X, P=P):
    """Share alpha sold to zero-NPV investors who pay `cash` (eq. 5.7)."""
    return cash / price(X, P)


c = claims(1.05)                                                                 # slides
 ↪ 68, 71
assert np.allclose([c["A"], c["D"], c["E"], c["rD"], c["rE"], c["rA"]], [1.75, 1,
 ↪ 0.75, 0.05, 0.40, 0.20])
a = alpha_equity(1.0)
assert np.isclose(a, 4 / 7) and np.isclose((1 - a) * price(X, P), 0.75)    # slide
 ↪ 69
c0 = claims(0.0)
assert c0["rD"] is None and np.isclose(c0["rE"], 0.20)                          # slide
 ↪ 72
c4, c5 = claims(0.42), claims(0.525)                                             # slides
 ↪ 78, 80
assert np.isclose(c4["E"], 1.35) and round(100 * c4["rE"], 1) == 24.4
assert np.isclose(c5["E"], 1.25) and np.isclose(c5["rE"], 0.26)
assert round(1.575 / (1 + c4["rE"]), 2) == 1.27                                 # slide 79
 ↪ fallacy: E 1.27
# slide 81: replicate the entrepreneur directly with stock (0, 2.8) and bond (1.05,
 ↪ 1.05)
for target in (np.array([0, 2.1]), 3 / 7 * X):
    xy = np.linalg.solve(np.array([[0, 1.05], [2.8, 1.05]]), target)        # rows:
 ↪ slump, boom
```

```
    assert np.isclose(xy.sum(), 0.75)                                   # both
     ↪ prices are 1


# ---------- 2A.8: MM I ----------
def arbitrage_step1(V1, V2, D1, E1, Rf, alpha, X):
    """Slide 87 portfolio when V2 > V1: short alpha of firm 2, buy alpha V2/V1 of
     ↪ firm 1's debt
    and equity. Returns (cost today, payoff per state). Firm 1's debt must be safe:
 ↪ Rf D1 <= min X."""
    k = alpha * V2 / V1
    cost = -alpha * V2 + k * D1 + k * E1
    payoff = -alpha * X + k * Rf * D1 + k * (X - Rf * D1)
    return cost, payoff


cost, pay = arbitrage_step1(1.75, 1.80, 1.0, 0.75, 1.05, 0.1, X)
assert abs(cost) < 1e-12 and (pay > 0).all() and np.allclose(pay, 0.1 * X * (1.80 /
 ↪ 1.75 - 1))


def put_call_value(K, X=X, P=P):
    """C(K) + K/(1+r_f) - P(K) (eq. 6.8); equals price(X) for every K."""
    call = price(np.maximum(X - K, 0), P)
    put = price(np.maximum(K - X, 0), P)
    return call + K * P.sum() - put


assert all(np.isclose(put_call_value(K), 1.75) for K in (0, 0.5, 1.05, 2.1, 3.15,
 ↪ 4.0))
assert np.isclose(claims(2.1)["D"], 2.1 * P.sum() - price(np.maximum(2.1 - X, 0), P))
 ↪ # eq. 6.6


# ---------- 2A.9-2A.10: MM II, WACC, yield ----------
def mm2_rE(rA, rD, D, E):
    """MM II (eq. 7.2): r_E = r_A + (D/E)(r_A - r_D); E > 0."""
    return rA + D / E * (rA - rD)


def yield_vs_rD(F, D, p_def, C):
    """(y, r_D) for one-period debt of face F and value D (eqs. 7.10-7.11); C
     ↪ recovery in default."""
    return F / D - 1, (F * (1 - p_def) + C * p_def) / D - 1


for F in np.linspace(0.01, 3.14, 300):                                  # both
 ↪ regions
    cl = claims(F)
    assert np.isclose(cl["rE"], mm2_rE(cl["rA"], cl["rD"], cl["D"], cl["E"]))
```

```
    assert np.isclose(cl["E"] / cl["A"] * cl["rE"] + cl["D"] / cl["A"] * cl["rD"],
     ↪ 0.20)  # WACC fixed
    if F > 1.05:
        assert np.isclose(cl["rE"], 0.40)                                    # flat
         ↪ region (7.6)
cr = claims(2.1)
y, rD = yield_vs_rD(2.1, cr["D"], 0.5, 1.05)
assert np.isclose(rD, cr["rD"]) and round(100 * y, 2) == 52.73 and round(100 * rD, 2)
 ↪ == 14.55 and y > rD
assert round(100 * mm2_rE(0.12, 0.08, 0.2, 0.8), 6) == 13 and round(100 *
 ↪ mm2_rE(0.12, 0.08, 1, 1), 6) == 16  # Ross p. 549
assert np.isclose(mm2_rE(0.15, 0.05, 15000, 15000), 0.25)                    # Berk p.
 ↪ 471
assert np.isclose((34500 - 6300) / 24000 - 1, 0.175)                         # Berk p.
 ↪ 469
print("L02 python: all checks pass")
```

What each block settles

Block	Settles	Note
`npv`, `irr_roots`, `tv_growing`	(2.3)–(2.6); two IRRs for the mine, none for $+100, -300, +250$; lending vs. borrowing	L02-02-cash-flows-npv-irr
`capm`, `unlever`, `relever`, `moments`	(1.4)–(1.12); the house's 8.25% / 1.25 and 12% / 2; the comparables' 10.56%	L02-01-present-value-capm-betas
`state_prices`, `price`	(4.1)–(4.6); 0.381 / 0.571 and $\pi = (0.6, 0.4)$; 0.357 / 0.595	L02-04-complete-markets-state-prices
`claims`, `alpha_equity`	every entrepreneur table (slides 68–80) and the slide-81 replication	L02-05-entrepreneur-example
`arbitrage_step1`, `put_call_value`	(6.1)–(6.8): zero cost, positive payoff; $V(K) = 1.75$ for every K	L02-06-mm1-arbitrage-put-call
`mm2_rE`, `yield_vs_rD`	(7.2) in both regions, WACC fixed at 20%, flat 40%, $y = 52.73\% > r_D = 14.55\%$	L02-07-mm2-wacc-cost-of-debt

Links

- Interactive version of `claims`: The State-Price Balance Sheet (`companions/companion-mm-state-prices.html`).
- Figures built from the same expressions: `figures/make_figures.py` (run `python -I Notas/L02/figures/make_figures.py`).
- Rules: rules/02 · rules/03.